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Influence of cutting edge preparation on tool wear during steel milling

Students: Lamice Denguir[a,b] , Xiaowen ZHU[a,c]

Advisors: Guillaume Fromentin[a], Aurélien Besnard[a], Abdelmajid Benamara[b]

a. Arts et Métiers ParisTech CER Cluny, France
b. ENIM, Monastir, Tunisia
c. Shanghai Jiao Tong University, Shanghai, China

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AUTEURS: Xiaowen ZHU / Lamice DENGUIR

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ENCADREMENT DU PE: Guillaume FROMENTIN / Aurélien BESNARD / Abdelmajid BENAMARA

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RESUME: The flank wear resistance of eight inserts is presented. One is the referential insert with no edge preparation; the seven other are prepared respectively by abrasive blasting, honing, abrasive flow machining, dragging, magneto-abrasive machining, brushing, and laser machining. Those SEAN 1203AFTN-M14 inserts are used to face mill the stainless steel AISI 304L and the hardened low alloy steel AISI 4140 QT. Starting with the comprehensive analysis of insert profiles, a best solution for precisely reconstructing the cutting edges is discussed, the process of down milling is explained with details including all the working parameters, and then flank wear trials results are revealed in wear charts as well. Then, basing on profile information, the relationship between insert preparation methods and tool life is interpreted. Finally, besides the geometry, we are interested in the correlation between cutting forces and flank wear behavior. So, a mechanical and statistical analysis is led to reveal laws relating cutting force components with VB.

MOTS CLES: WEAR / INSERT EDGE PREPARATION / STEEL MILLING / PCA

PARTIE A REMPLIR PAR LE PROFESSEUR RESPONSABLE DU PROJET		
ACCESSIBILITE DE CE RAPPORT (entourer la mention choisie) :		
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Date :	Nom du signataire :	Signature :

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Finally, we are deeply grateful to our parents, who did sacrifices to make us able to achieve such a project.

Contents

N.B: Appendix is attached in DVD form

Introduction

With the development of various cutting methods, the mechanical preparation of cutting tools gains more and more in importance. Proved by the researchers worldwide, it brings two advantages for commercialized cutting inserts: One is to generate a high quality, defect-free cutting tool surface for succeeding coating operations. The other is to delicately control a clearly defined micro-geometry of the cutting edge. Due to these improvements, the tool life is expected to be extended, and so the economic aspect is improved.

With the partnership of the International Academy for Production Engineering (CIRP), this project makes a part of an international Robin test. It is a research method which launches one research subject in several laboratories simultaneously, definitely specifies the same experimental conditions, and respects the same functional specifications in order to compare results. Its advantage is obviously—to maximize and guarantee the accuracy, reliability, as well as repeatability of the research.

Our objective is to find the influence of cutting edge preparation method on tool wear while steel milling. This report aims to compare the influence of different edge preparation methods on the tool wear by means of rough face milling process.

The edge preparation techniques studied include abrasive blasting, abrasive flow machining, brushing, dragging, honing, laser machining and magneto-abrasive machining. Additionally, we have untreated inserts as reference. These techniques are applied on standard TiN coating cutting inserts SEAN1203AFTN-M14 with the tool holder (Dc=50mm), both fabricated by SECO. According to unified feedback and the sketched work plan, we are invited to conduct cutting tests in face milling of two materials: the hardened steel AISI 4140 QT (42CrMo4 quenched and tempered) and the stainless steel AISI 304L (X2CrNi19-11)

In this report, firstly, a brief literature review about edges preparation will be demonstrated in Chapter 1. Then, characterization of insert geometry, workpiece materials and design of the rough face milling process will be revealed in Chapter 2. After that, the results of tool life test will be demonstrated by comprehensive wear charts in terms of both maximum flank wear $V_{B\,max}$ and average flank wear $V_{B\,av}$ in Chapter 3, on what a qualitative analysis is based, engaging wear observation and chips measuring. Moreover, for quantitative analysis, in Chapter 4, principal component analysis (PCA) will be applied in order to exploit the correlations among cutting forces, working angles, inserts and chips surface state with wear results. This analysis will lead to a statistical model for predicting flank wear. Finally, a comprehensive conclusion will be given and our perspectives will be raised.

A short view on CIRP

CIRP was founded in 1951 with the aim to address scientifically, through international co-operation, issues related to modern production science and technology. The International Academy for Production Engineering takes its abbreviated name from the French acronym of College International pour la Recherche en Productique (CIRP) and includes some 600 members from 50 countries. The number of members is intentionally kept limited, so as to facilitate informal scientific information exchange and personal contacts.

CIRP includes some 150 Associate members well known scientists, with high potential, elected typically for a period of three years with the possibility of renewal. A number of Associate members eventually become Fellows. Some Associate members may also belong to fields related to Manufacturing.

CIRP, although an academic organization, encourages the participation of industry in its activities. There are some 150 corporate members who follow the research work of the academic members of CIRP, and very often contribute to the information exchange within CIRP by presenting their views on industrial needs and perspectives.

Young Research Affiliates and some Invited members, particularly from countries not yet involved in CIRP, are also included in the CIRP community.

In a recent development, there is work under way to establish a CIRP Network of young scientists active in manufacturing research.

CIRP aims in general at:

- Promoting scientific research, related to manufacturing processes, production equipment and automation, manufacturing systems and product design and manufacturing
- Promoting cooperative research among the members of the Academy and creating opportunities for informal contacts among CIRP members at large.
- Promoting the industrial application of the fundamental research work and simultaneously receiving feedback from industry, related to industrial needs and their evolution.
- Organizing an annual General Assembly with keynote and paper sessions and meetings of the Scientific and Technical Committees, publishing papers, reports, annals and other technical information, organizing and sponsoring international conferences.

CIRP is organized along the lines of a number of Scientific Technical Committees (STCs) and Working Groups (WGs), covering many areas. CIRP organizes annually a General Assembly and the so called January Meetings.

The main publications of CIRP are the CIRP Annals under ISI standards with two volumes; Volume I, with refereed papers presented in the GA by Fellows, Associate, Corporate and Invited members and Volume II with refereed keynote papers. There are also CIRP proceedings, including round table discussions, technical reports, special issues, reports and internal communications, proceedings of CIRP conferences, dictionaries of production engineering etc. A Newsletter is also published twice a year. Currently the CIRP Annals are published by Elsevier, while Springer Verlag publishes the Dictionaries of Production Engineering. There are under development one or more journals, complementing the work published in the CIRP Annals.

- The International Academy for Production Engineering
- College International pour la Recherche en Productique
- Internationale Akaddemie für Produktionstechnik

Fig.0.1. CIRP logo [1]

Chapter 1: Literature review

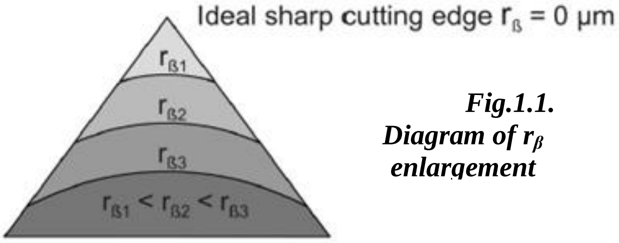


Fig.1.1. Diagram of r_β enlargement

The mechanical preparation of cutting tools gains more and more importance. There are two advances brought by preparation: Firstly, it generates a high quality, defect-free cutting tool surface for succeeding coating operations. Secondly, the generation of a clearly defined micro-geometry of cutting edge. In our case, a sharp angle exists between chamfer and flank in our cutting inserts. This sharpness can possibly lead to the easily cracking on the cutting edge during the interaction between insert and workpiece. Different techniques are used to reduce the angle and create tiny radii marked as r_β (Fig.1.1).

The value of r_β depends on the methods used. In this research, we focus on the tool life of one referential insert without preparation process along with 7 inserts manufactured by the following techniques (Fig.1.2).

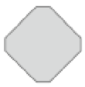
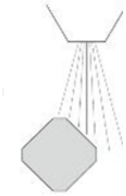
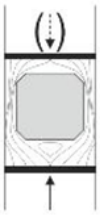
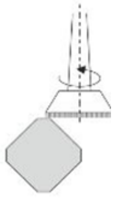
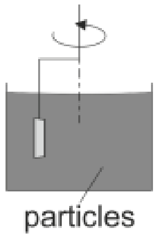
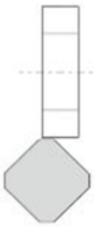
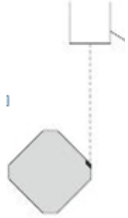
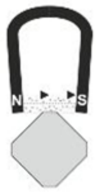
Untreated (reference)	Abrasive blasting	Abrasive flow machining	Brushing	Dragging	Honing	Laser machining	Magneto-abrasive machining
							
0374, 0375	**0429, 0430**	**0025, 0026**	**0575, 0576**	**0244**	**0699, 0700**	**0639, 0640**	**0532, 0533**

Fig.1.2. Referential insert and 7 preparation techniques with their corresponding inserts number

Preparation methods refer to manufacturing processes which modify the shape and surface texture of inserts. A wide range of such process is employed to produce high quality parts to high accuracy and to close tolerances.

In the next paragraphs, basing on figures elaborated by K.D. Bouzakis, N. Michailidis, G. Skordaris and E. Bouzakis [2], edge preparation methods used in our study will be explained.

1.1 Abrasive blasting

A nozzle sprays small particles of fine sands on the insert.

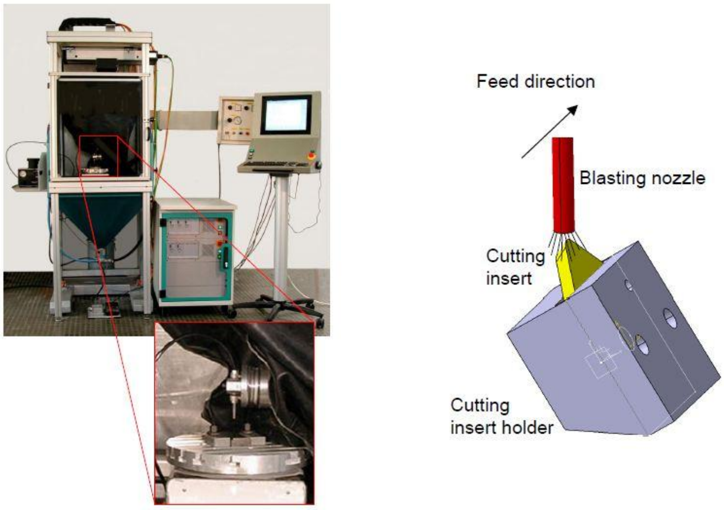


Fig.1.3. Abrasive blasting → process

1.2 Abrasive flow machining

G. F. Schrader, A. K. Elshennawy [3] defined this method as mixed small particles with liquid medium modifying insert surface and obtaining controlled radii. One-way systems flow the media through the workpiece, then it exits from the part. In two-way flow, two vertically opposed cylinders flow the abrasive media back and forth.

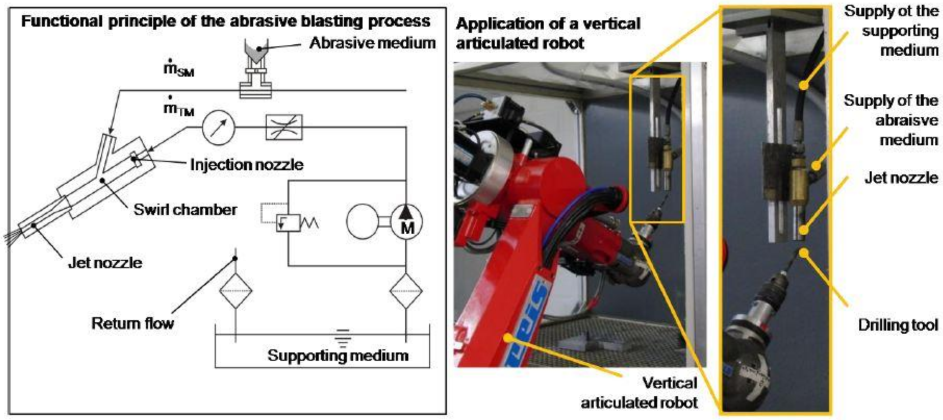


Fig.1.4. Abrasive flow machining process

1.3 Brushing

A brush rotates and contacts the insert directly to generate controlled radii by friction.

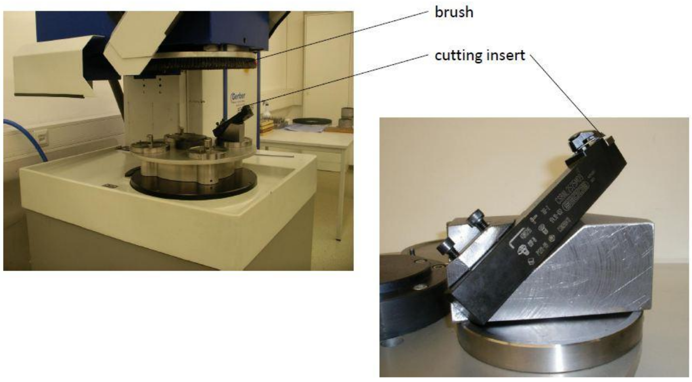


Fig.1.5. Brushing process

1.4 Dragging

Insert is submerged under a medium consisting of particles. By rotating the insert, each edge of insert is treated evenly.

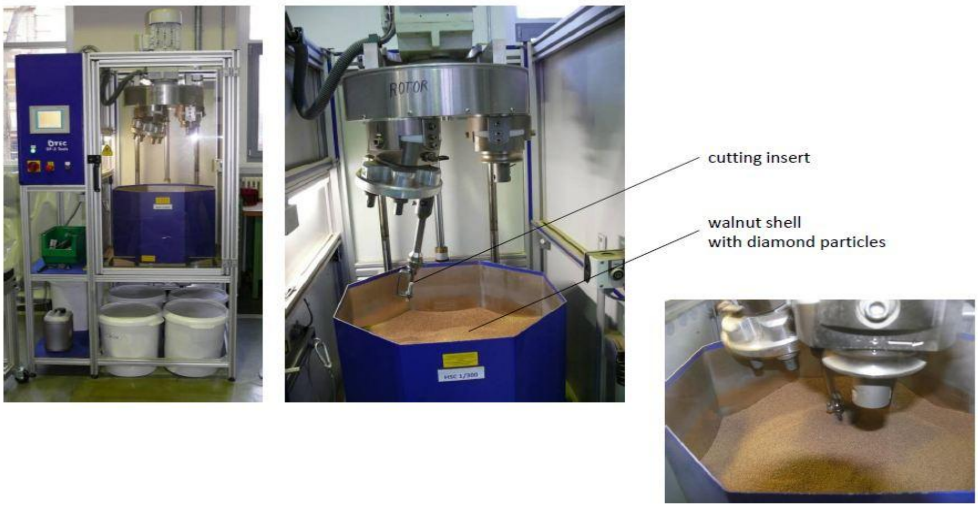


Fig.1.6. Drugging process

1.5 Honing

A wheel rubs against the insert at a low speed and produce rolling friction to increase the r_β.

1.6 Laser machining

Using laser beams to ablate the inset edge. The high-energy beams remove materials by extreme heat.

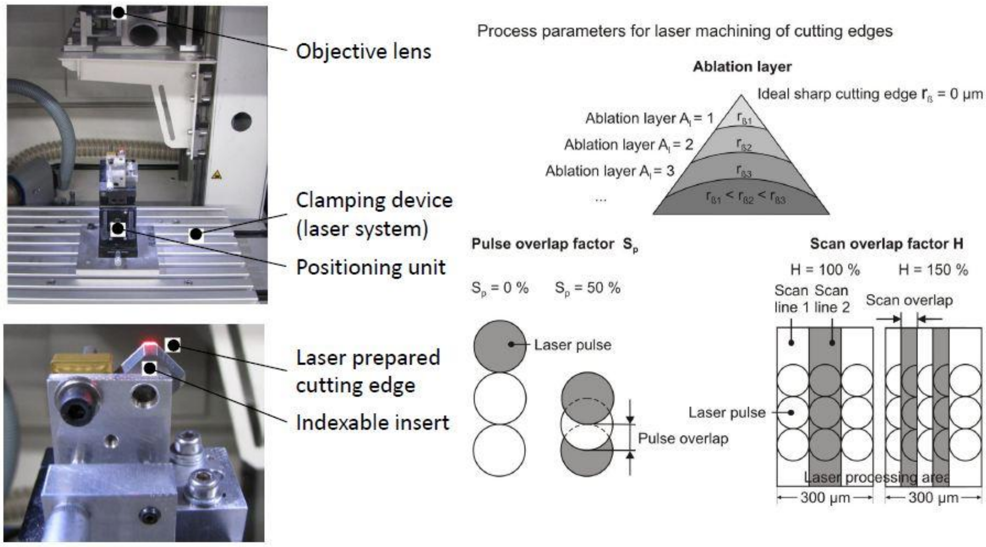


Fig.1.7. Laser machining process

1.7 Magneto-abrasive machining

According to B. Karpuschewski, O. Byelyayev and V.S. Maiboroda [4], the magnetic system consists of two ring-shaped pole shoes in coaxial setting. These pole shoes create the working zone, which is filled with magneto-abrasive powder. According to CIRP annals [5], magneto-abrasive machining enables the generation of a reproducible, consistent rounding of the tool cutting edges and additionally improves the quality of the corner edges as well as the quality of the tool surfaces that form the cutting edge.

Conclusion

In this first chapter, in a brief literature review, the different edges preparation methods applied to the inserts afforded to us for our study are explained. In the next chapter, the parameters characterizing the Robin test done in this project will be detailed as well.

Chapter 2: Parameters of the study

This chapter explores the properties of inserts as well as workpieces materials. Initial geometry of inserts, profile measurement methods and compositions of milled materials will be revealed.

2.1 Inserts characterization

In this project, the selected cemented carbide inserts, SEAN 1203AFTN-M14 coated with F40M (fig.2.1) are studied. The differences among the inserts are due to different preparation processes. Seven types of pre-treatment are listed as following: abrasive blasting, abrasive flow machining, brushing, dragging, honing, laser machining and magneto-abrasive machining

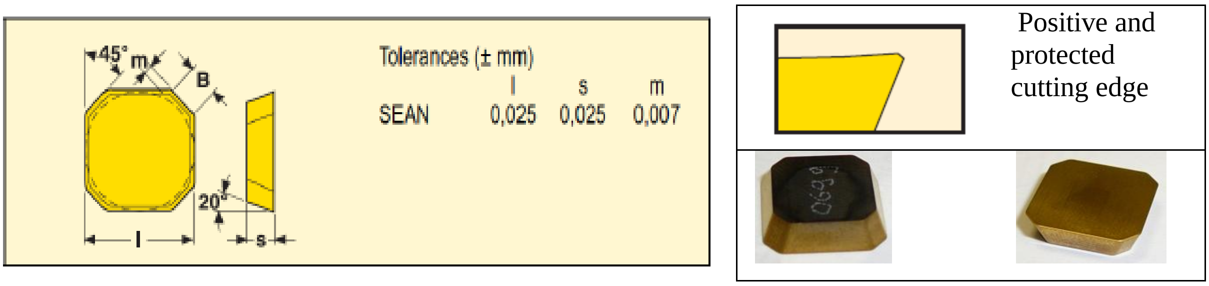


Fig.2.1. Insert: SEAN 1203AFTN-M14 coated with F40M

Cemented carbide is an alloy of tungsten carbide (WC) and cobalt (Co). Cubic carbides like tantalum carbide (TaC), titanium carbide (TiC) and niobium carbide (NbC) can also be added. Tungsten carbide is the main component and gives the hardness. Cobalt is the binder phase and gives the toughness. Cubic carbides are added in order to affect properties like hardness, deformation resistance and chemical wear resistance.

Our inserts are coated with the coating material TiN by PVD (Physical Vapor Deposition) technique which is an extremely hard ceramic material. The coating improves the wear resistance of the grade. PVD-coated grades are recommended for applications with low feed rate where high edge toughness is required. PVD-coated grades are suitable for applications with low to intermediate cutting speed.

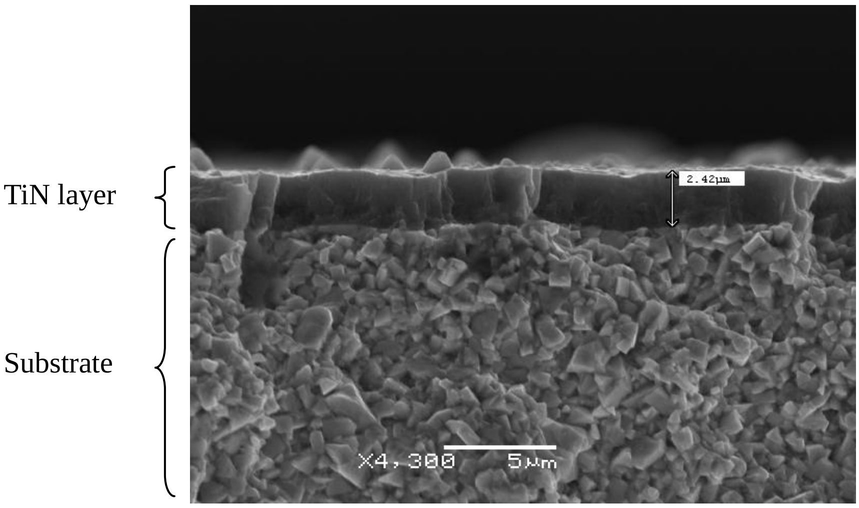


Fig.2.2. Cross-sectional SEM micrograph showing the carbide substrate and its supporting TiN layer.

This TiN grade, known as F40M, is a basic grade in the commercial series of insert products. It is suitable for fine to medium rough milling. It is also the first choice for milling with small feeds and or low cutting speeds. Excellent for milling when there is a risk of vibrations and when coolant is used, recommended for machining superalloys.

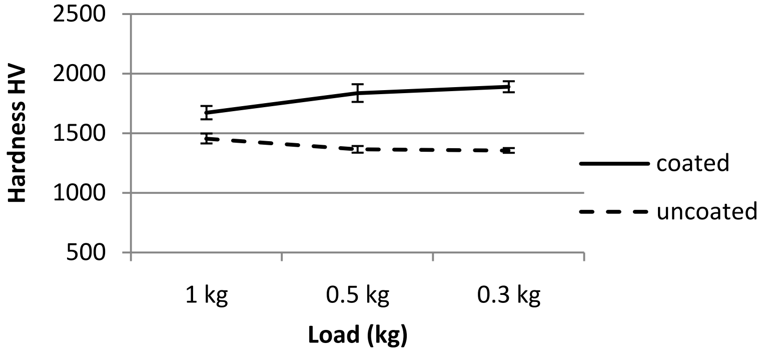


Fig.2.3 Micro-hardness of insert (insert sample 0639)

Fig.2.3 demonstrates the effect of the (Ti, Al) N – TiN layer on one of our insert obtained by instrumented indentation. The coating layer reinforces at least 15% of the hardness on the tool surface. With the variation of the load on the indentations, the hardness of uncoated face of insert varies in a reasonable range while the hardness of coated face shows an increase with the decrease of load. It is due to the destruction of the coating layer.

2.1.1 Geometric parameters

For better understanding the geometry of the inserts, all the concerning faces and edges are indicated in Fig.2.3. At the same time, all the geometry angles of the inserts are marked in Fig.2.4.

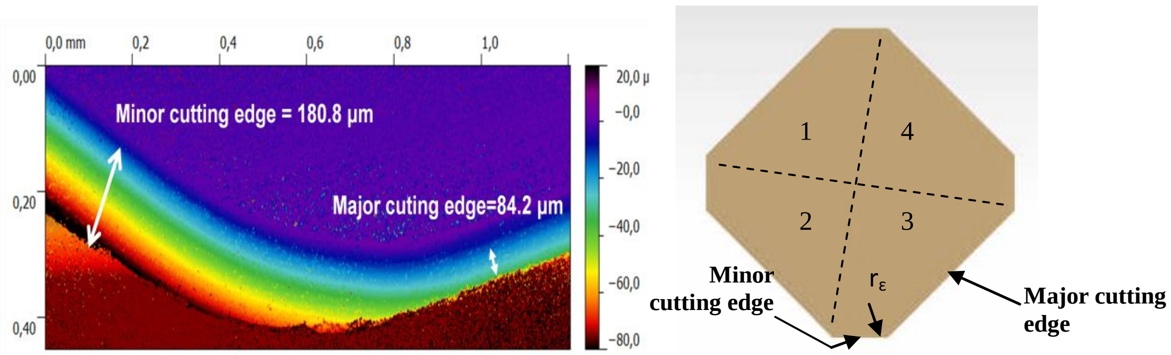


Fig.2.4. Definition of different faces and cutting edges of insert (Right: cutting edges projected by Wyko; Left: the repartition of the cutting edges)

During our study, each insert will be down milling twice to guarantee the repeatability of our tests. That means among the 4 edges demonstrated in Fig.2.3 (Each edge consists of a minor cutting edge and a major cutting edge), only two of them will be selected. It is not necessary for these two edges to be in opposite position. The most important is to insure they are in the best condition compared to the others.

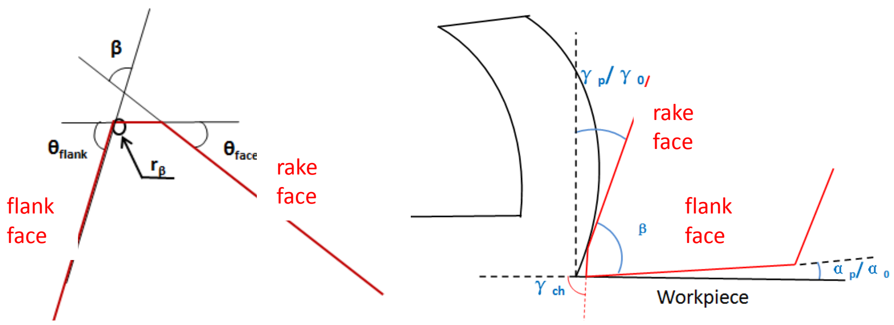


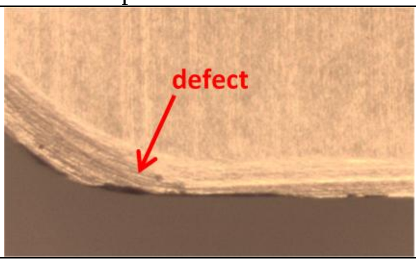
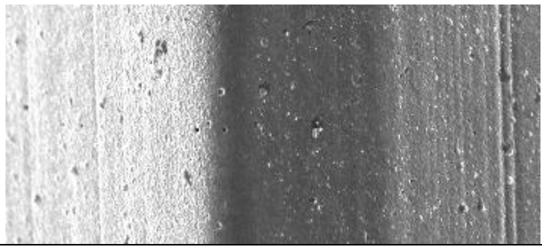
Fig.2.5. Definition of profile angles and working angles

It is worth to notice that the right diagram of fig.2.4 illustrates not only the working condition of minor cutting edge but also the major cutting edge. In the case of minor cutting edge, according to the tool assembly mode, γ_p is 20°. However, in the case of major cutting edge, γ_o is 12°.

θ_{face}	Angle between chamfer and cutting face
θ_{flank}	Angle between chamfer and flank
β	Angle between face and flank
r_β	Radius between flank and chamfer
γ_0	Angle between face and vertical plane for major cutting edge
γ_p	Angle between face and vertical plane for minor cutting edge
α_0	Angle between flank and workpiece for major cutting edge
α_p	Angle between flank and workpiece for minor cutting edge
γ_{ch}	Angle between chamfer and workpiece

Tab.2.1. Terminology

Every insert is characterized by a different morphology due to preparation method (Tab.2.2). By comparing the status of the edges, we choose 2 edges with best qualities to carry out the down milling tests for each insert (cf. Appendix 2_1)

	Optical observation	Morphology
Cutting edge		

Tab.2.2. Observation of morphology

Preparation	**Inserts references**	**Edges selected**	
reference	0374	1	3
	0375	1	3
abrasive blasting	0429	2	4
	0430	2	4
abrasive flow machining	0025	2	4
	0026	1	3
brushing	0575	2	4
brushing	0576	1	3
laser machining	0639	1	4
	0640	2	4
dragging	0244	2	3
honing	0699	2	3
	0700	2	4
magneto-abrasive machining	0532	1	4
	0533	2	4

Tab.2.3. Results of edge selection

2.1.2 Methods of geometric measurement

For profile measurement, there are multiple methods corresponding to our case. Generally, they can be divided into 2 classes, contact measurement and non-contact measurement. It is important to select the right instrument for the application, since each instrument has benefits and drawbacks. It is required to choose a suitable apparatus. In our study, three methods are available which are Optical profiler, mechanical profiler and Scanning electron microscope.

a) Optical profiler (Without contact)

Wyko NT1100 optical 3D profiler manual metrology system uses non-contact interferometry for highly accurate. It gives also roughness analysis, like linear roughness or statistic surface roughness.

During the measuring process, we chose white light vertical scanning interferometry with magnification of x5, x10 and x20. Its vertical measurement range reaches to 0.1 nm to 1 mm and the vertical resolution is smaller than 1Å Ra. Limited to the field size and depth of Wyko, we have to detect one profile twice with two directions; one direction for detecting cutting face and chamfer and the other for detecting chamfer and flank. By combining the information of two images (Fig.2.6), the complete profile is obtained (Fig.2.7).

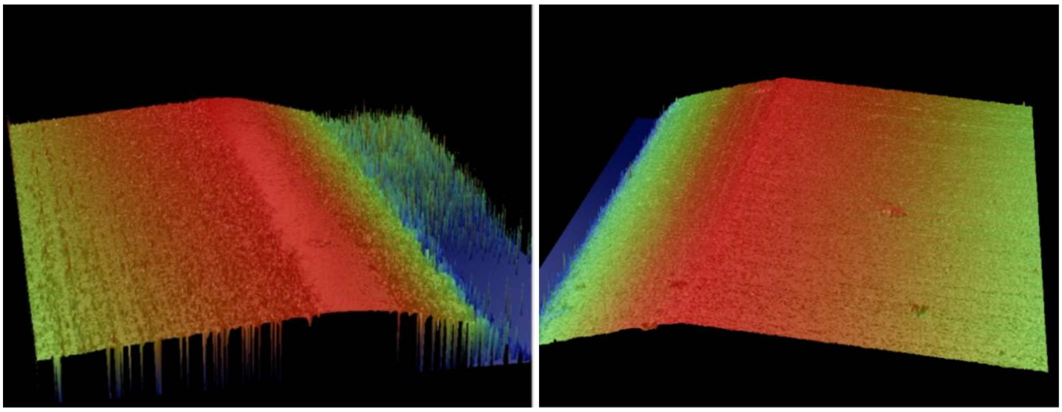


Fig.2.6. Left: Flank- chamfer by Wyko; Right: Chamfer-cutting face by Wyko

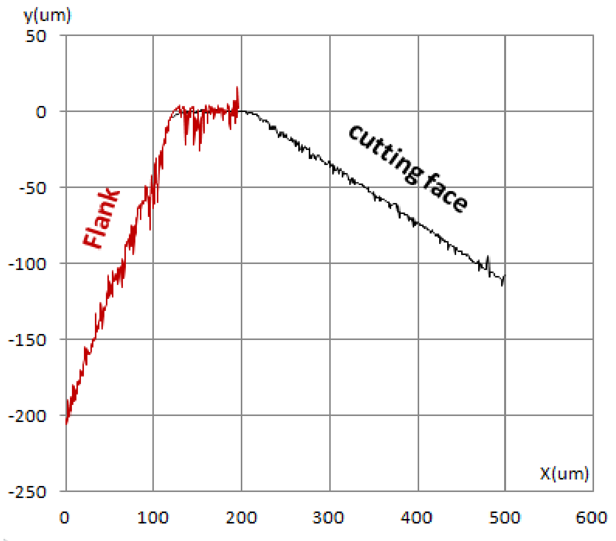


Fig.2.7. Combined profile of Wyko

b) Stylet profiler (With contact)

As a type of stylet profiler, Somicronic SurfaScan 3D is possibly the simplest equipment for surface characterization. By recording the distance traversed on the sample surface on the abscissa and the height measurement on the ordinate, a two dimensional map of the surface can be constructed. The principle of this instrument is to use the variation of the mechanical motion during scanning process.

We have used a stylet of serial SP265 characterized with a radius of 50 μm and minimum detective angle of 30°.

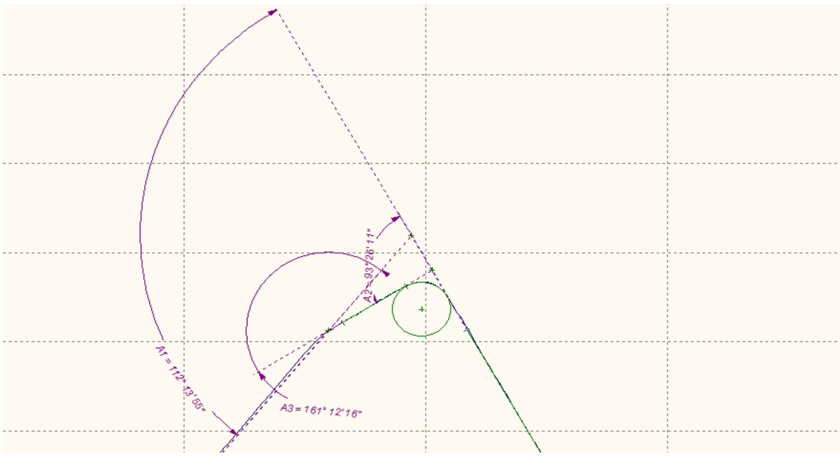


Fig.2.8. Profile reconstructed by SurfaScan

c) Scanning Electron Microscope (Without contact)

Scanning electron microscope (SEM) produces secondary electrons, back-scattered electrons, characteristic X-rays, light, specimen current and transmitted electrons. When the signals impinge the surface of the sample, interactions of the electron beam with atoms at or near the surface will take place. Since the electron beam is very narrow, SEM micrographs have a large depth of field yielding a characteristic three-dimensional appearance useful for understanding the surface structure of a sample.

With the aid of the software MEX 5.0, the 3D profile of cutting edge can be reconstructed. During the measurement, it takes 3 scanning photos to combine a complete 3D edge profile. The tilt angles between these three photos are always kept at 7° both to the left and to the right.

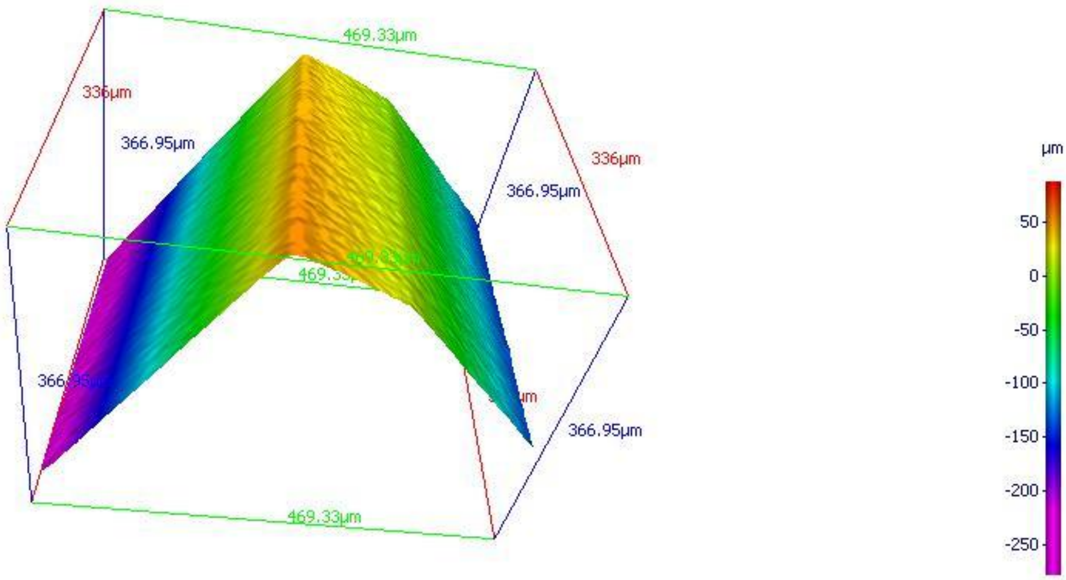


Fig.2.9. Profile reconstructed by SEM

We favor the results gained by SurfaScan for the reasons below:

i. For r_β, the values have great discordant results among three methods. r_β values detected by

Wyko are evidently incorrect because r_β cannot be otherwise but increase after process of abrasive blasting or honing. However, Wyko gives a smaller r_β of honing insert than that of reference inserts which is not at variance with objective reality. That is due to the poor qualities of flank-chamfer images which have too many defaults in the images. They are caused by the high slopes between flanks and chamfers that cause relatively big depth variation during interferometry. The same reason leads to the inaccuracy of θ_{flank} angle and negative α angle.

ii. For SurfaScan and SEM, r_β evaluated by SurfaScan is more than twice as big as that by SEM. But it should be noticed that tab.2.4 and tab.2.5 only show the average values of parameters. When we look into the initial data (see attachment 1< profile geometry/sheet: complete data>), it appears that SurfaScan has a perfect repeatability and stability in measuring radius r_β, while the values of SEM have inexplicable fluctuation and a high deviation compared with that of SurfaScan (Fig.2.9).

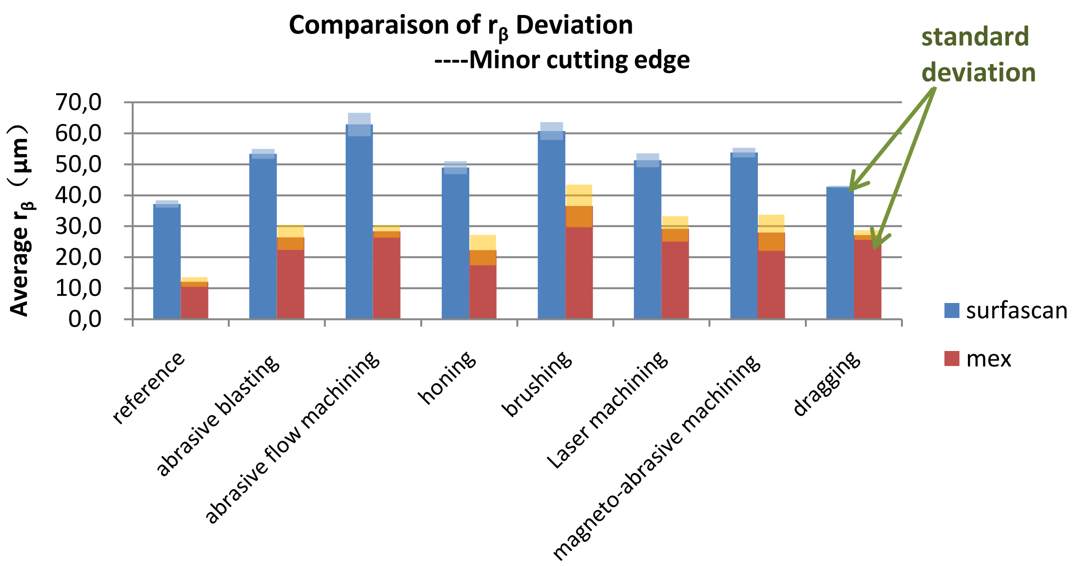


Fig.2.10.a. A comparison of r_β deviation in minor cutting edge (cf. Appendix 2_2)

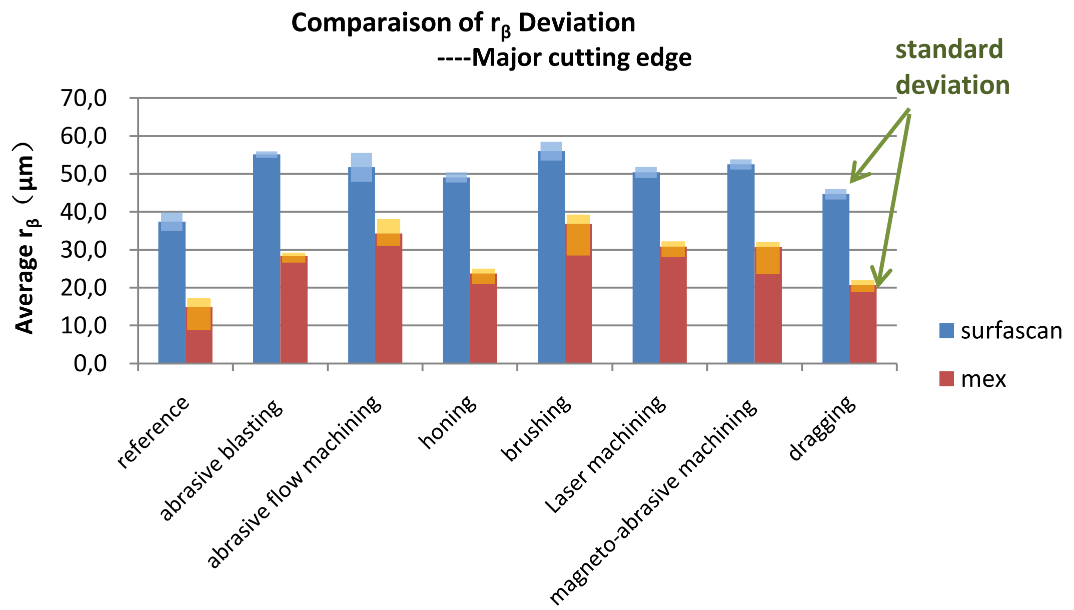


Fig.2.10.b. A comparison of r_β deviation in major cutting edge (cf. Appendix 2_2)

iii. Moreover, for SurfaScan, the radius is obtained by fitting a circle around the corner between chamfer and cutting face. Since the profile of SurfaScan is relatively smooth, usually the circle is well fitted. But for SEM, we have tried two types of software, MeX5.0 and Gwyddion 2.26, but we still have difficulty to fit a circle since the corners in the image of SEM are sharper. Imperfect fitting like the red part in Fig.2.11 always exists leading to errors in radius values.

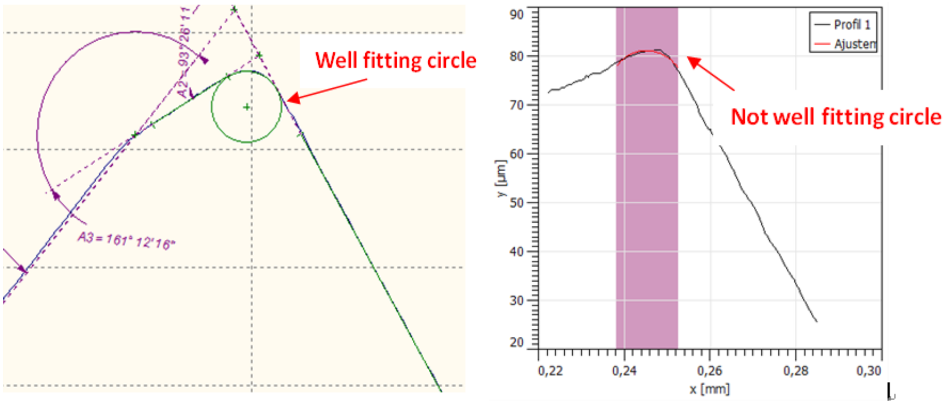


Fig.2.11. Left: radius measurement r_β by SurfaScan; Right: radius measurement r_β by SEM using software Gwyddion

2.1.3 Geometric results

Thus, after comparison, we have accepted the results obtained by SurfaScan as geometry information of inserts (Tab.2.4)

Minor cutting edge	reference	abrasive blasting	abrasive flow machining	brushing	dragging	honing	laser machining	magneto-abrasive machining
r_β (µm)	37.2	53.4	62.9	60.8	42.8	48.9	51.3	53.8
θ_{face} °	18.6	18.8	18.9	17.8	18.8	18.4	18.6	20.8
θ_{flank}°	93.9	94.1	93.6	94.7	93.4	93.4	93.7	93.5
β°	67.4	67.1	67.5	67.6	67.7	67.7	67.7	67.7
α_p°	2.5	2.9	2.5	2.4	2.3	2.3	2.3	2.3
length of chamfer (µm)	183.8	182.6	178.1	174.7	181.1	187.3	181.9	177.4
Ra	0.20	0.39	0.45	0.29	0.3	0.3	0.2	0.4
Rt	2.04	3.59	3.90	3.02	2.5	2.9	1.8	4.2

Major cutting edge	reference	abrasive blasting	abrasive flow machining	brushing	dragging	honing	laser machining	magneto-abrasive machining
r_β (µm)	37.4	55.1	51.8	56.0	44.6	49.1	50.4	52.5
θ_{face} °	17.8	17.2	17.5	18.1	19.0	18.1	18.4	18.5
θ_{flank}°	87.0	87.7	87.8	87.1	87.9	87.6	86.9	86.2
β°	74.8	74.8	74.6	75.0	74.0	74.5	74.7	75.2
α_0°	60.2	60.8	3.4	3.0	4.0	3.5	3.3	2.8
length of chamfer (µm)	88.3	89.9	80.8	80.4	76.6	88.1	84.0	85.6
Ra	0.13	0.21	0.29	0.25	0.3	0.4	0.4	0.4
Rt	1.20	2.06	2.43	2.15	2.5	3.4	3.3	4.1

Tab.2.4. Inserts geometry data

2.2 Machined materials characterization

2.2.1 AISI 304L (X2CrNi19-11)

Down milling tests are done using AISI 304L sample workpieces. Classified according to their crystalline structure, 304L belongs to the 300 series which have an austenitic crystalline structure, characterized with a FCC crystal structure. It can retain an austenitic structure at all temperatures from the cryogenic region to the melting point of the alloy, thanks to its max 0.15 % carbon, min 16 % Cr and sufficient Ni and Mn. 304L is the low-carbon version which is often applied to avoid corrosion in certain corrosive media after it is welded or otherwise heated at temperatures between 430 and 860°C. According to our material provider UGITECH, this stainless steel has the mechanical properties and chemical composition shown in the tables below:

Properties	**AISI 304l**	**Properties**	**AISI 304l**
Dimensions	50x50x500	Z [%]	77
R_m [MPa]	591	Thermal Conductivity [W m $^{-1}$°C $^{-1}$]	16.2
$R_{p0,2}$ [MPa]	348	Thermal Expansion Coefficient [°C $^{-1}$]	17.3 x 10^{-6}
A [%]	55	Electrical Resistivity [W m]	7.2 x 10^{-7}

Tab.2.5. Mechanical properties of AISI 304L

For hardness, several series of indentation have been conducted on the surface its average Brinell hardness is about 190.25 HB and average micro-hardness is nearly 227.42 HV (Fig.2.12 and Fig.2.13).

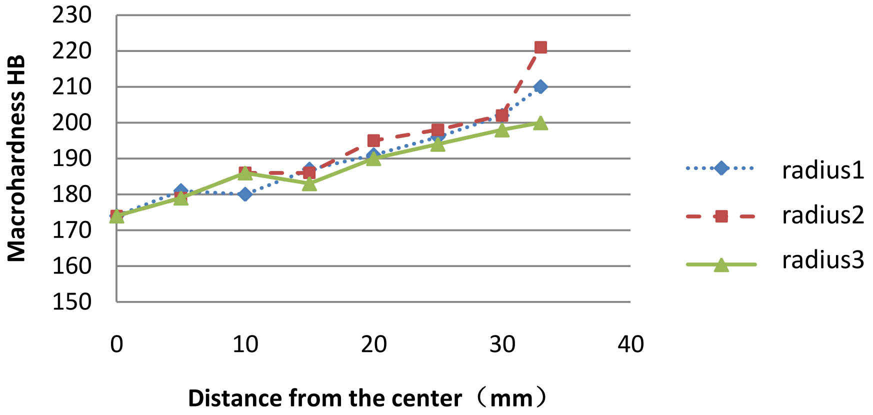


Fig.2.12. Brinell hardness of AISI 304L (Average = 190.25 HB)

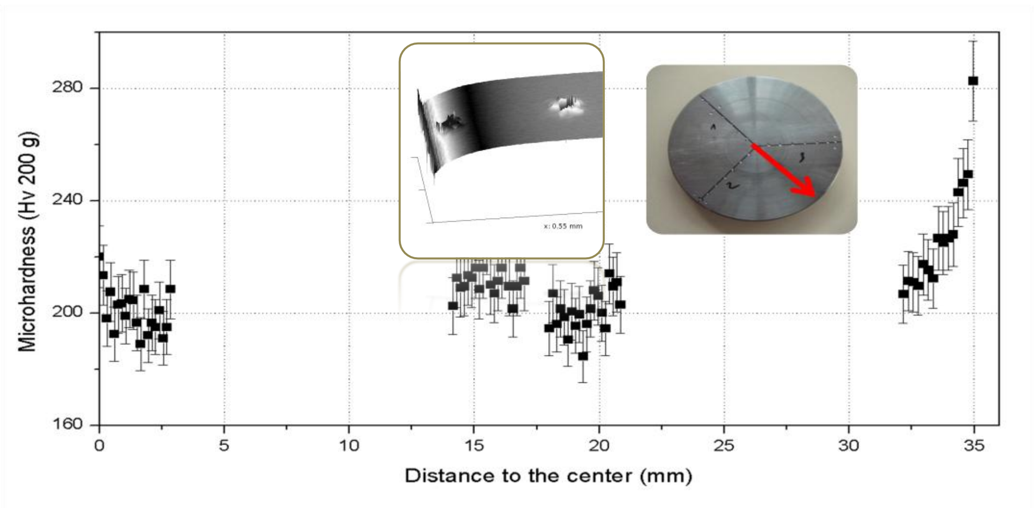


Fig.2.13. Micro-hardness of AISI 304L

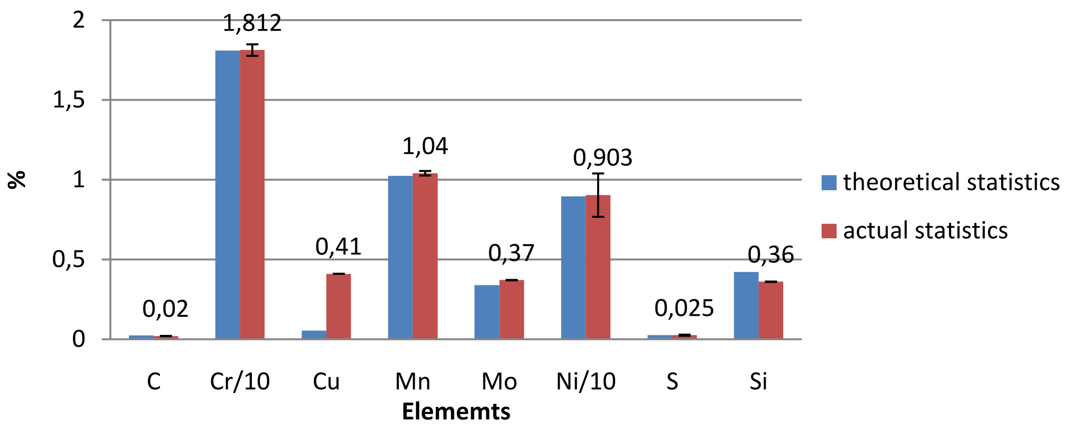


Fig.2.14. Elements analysis by spectrometry (in weight %)

Fig.2.14 above verifies the actual elements contained in our workpiece with the theoretical values announced by the fabricant. Except an evident excessive continent of Cu, other main elements stratify the material requirement. The extremely low carbon continent makes it belong to the family of mild steel and impossible to form carbide. Since Cr% exceed 10.5%, 304L stainless steel. S% is a important index because it improve machinability by forming low shear strength inclusion in the cutting zoon and allow the steel to deform more easily.

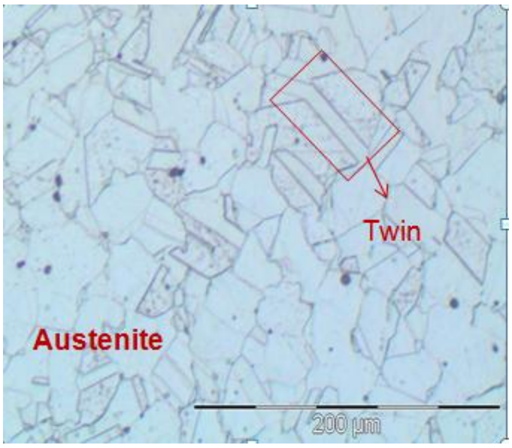


Fig.2.15. Microstructure of 304L (left x20, right x100)

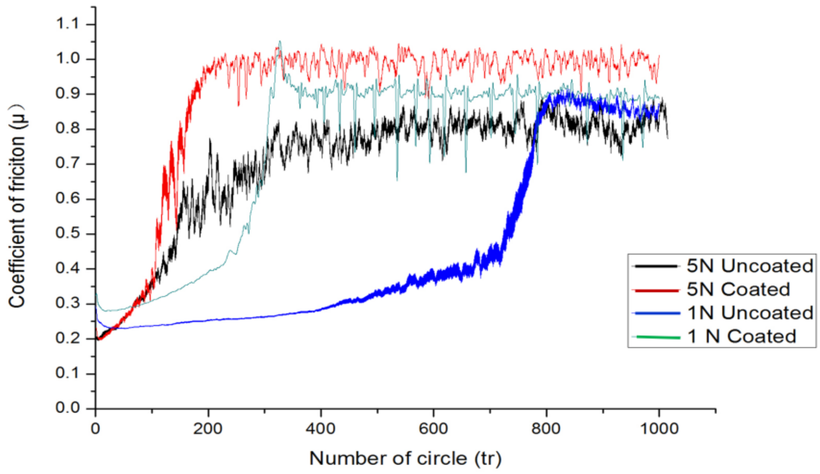


Fig.2.16. Tribology between 304L and insert (results of trials done with CSM apparatus)

Observation of the microstructure of 304L shows clear characteristics of austenite: polygonal ferrite grain with nano-scale twins. For explanation, a mixed structure of austenite grains embedded with twin bundles is induced by dynamic plastic deformation following by its thermal annealing. Twins are able to elevate alloy strength, while the ductility from the re-crystallized grains.

Stable state	1 N Coated surface	1 N Uncoated surface	5 N Coated surface	5 N Uncoated surface
Coefficient of friction	0.89	0.57	0.99	0.84

Tab.2.6. Average coefficient of friction (304L) in the stable state

Experiments of tribology have been conducted to characterize the coefficient of friction between insert and 304L by a mode of pion-on-disc. Both faces with coating and without coating of the inserts were tested as disc while pion (slider) was made of 304L with a radius of 6 mm.

Friction curves are showed in Fig.2.16, two states are observed: the initial transient state and the stable state. Under the same load, the transient stage of the uncoated surface last longer than that of the coated face. When changing the load on the same surface condition, it appears that the coefficient of friction increases with the increasing of load 1N to 5N. In addition the curves reach stable state faster under the applied load of 5N.

The microscope revealed that the coating was not worn out by friction. Instead, only the pion made of 304L stainless steel (slider) transferred to the surface of TiN coating surface. That explains the evolution of curve: In general, friction is believed to result from 3 components [6]; adhesion, ploughing and asperity deformation. At the beginning, the softer 304L was ploughed by hard asperity of TiN coating. With increasing slide distance, 304L gradually accumulated discontinuously on the coating surface. As sliding continued, further contact occurred between insert and 304L and strong adhesion junction turned out to form at the region of real contact between 304L sliders and its remains on the coating. Thus the coefficient of 0.89 and 0.99 represent the friction between stainless steel while the coefficient of 0.2～0.3 represents the friction between stainless steel and hard carbide insert.

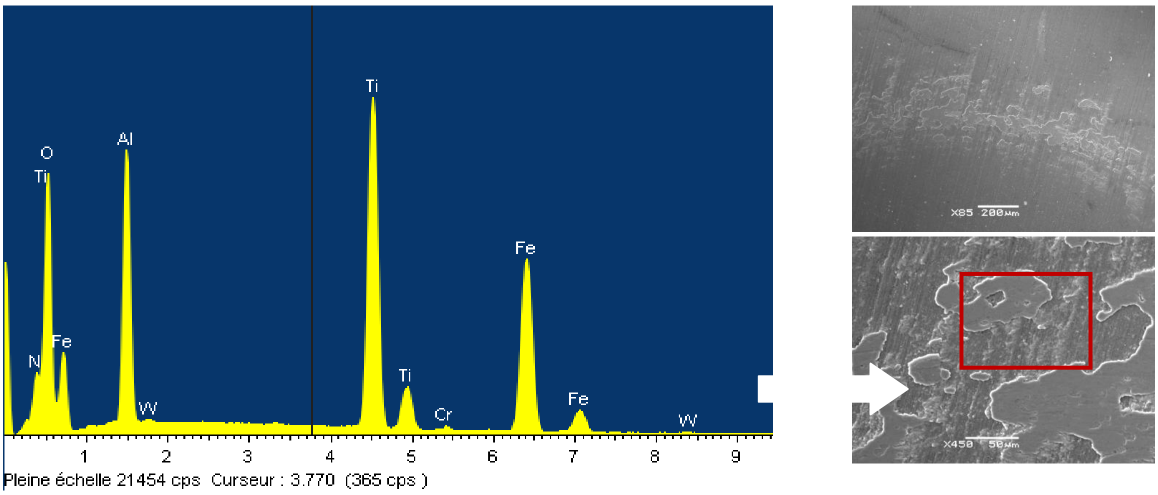


Fig.2.17. EDS analysis of friction path

2.2.2 AISI 4140 QT(42CrMo4 QT)

Quenched-and-tempered 4140 is usually manufactured from the standard low-alloy chromium-molybdenum steel. This material finds application where there is a requirement for elevated strength in combination with a defined and high level of toughness. The QT treatment refers to a process of hardening followed by tempering in the highest possible temperature. It results in grain refinement and transformation of plate-shaped (perlite) and banded (bainite) carbides into a dispersion of spheroid carbides. The following table gives out the mechanical properties of AISI 4140 QT (Tab.2.7).

Properties	AISI 4140 QT
Dimensions	50x50x500
Rm [MPa]	860～1060
Rp0,2 [MPa]	730
A [%]	14
Z[%]	45

Tab.2.7. Mechanical properties of AISI 4140 QT

AISI 4140 QT has a superior hardness compared to other alloys steels. Its Brinell hardness is about 298.25 HB and its micro-hardness is about 311.10 HV (Fig.2.18 & Fig.2.19).

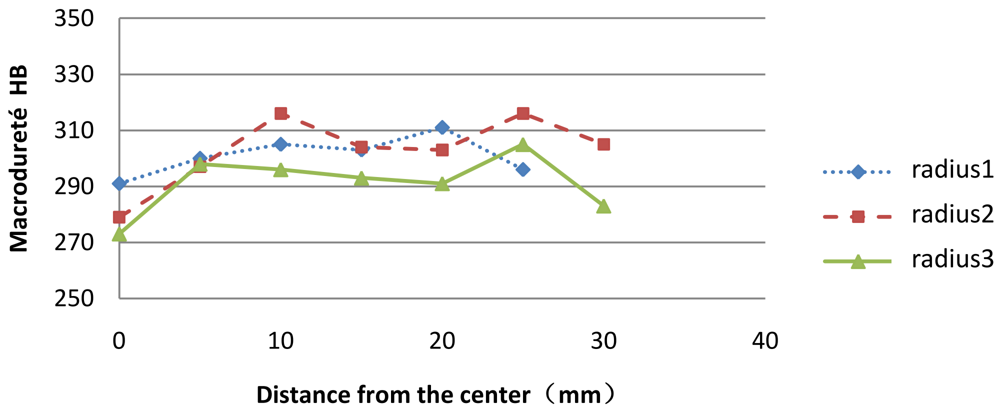


Fig.2.18. Brinell hardness of AISI 4140 QT (Average = 298.25 HB)

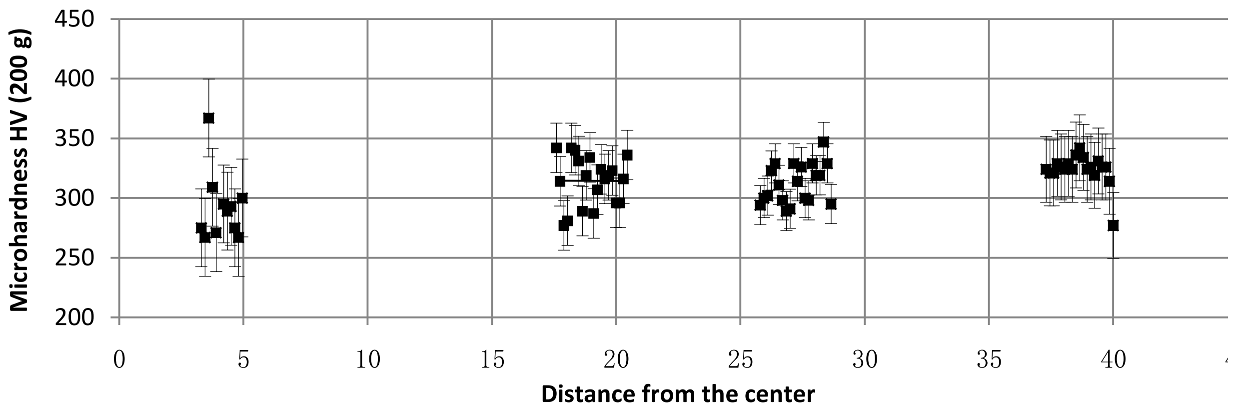


Fig.2.19. Micro-hardness of AISI 4140 QT

Stable state	1 N Coated surface	1 N Uncoated surface	5 N Coated surface	5 N Uncoated surface
Coefficient of friction	0.69	0.68	0.52	0.55

Tab.2.8. Average coefficient of friction (4140 QT/insert) in the stable state

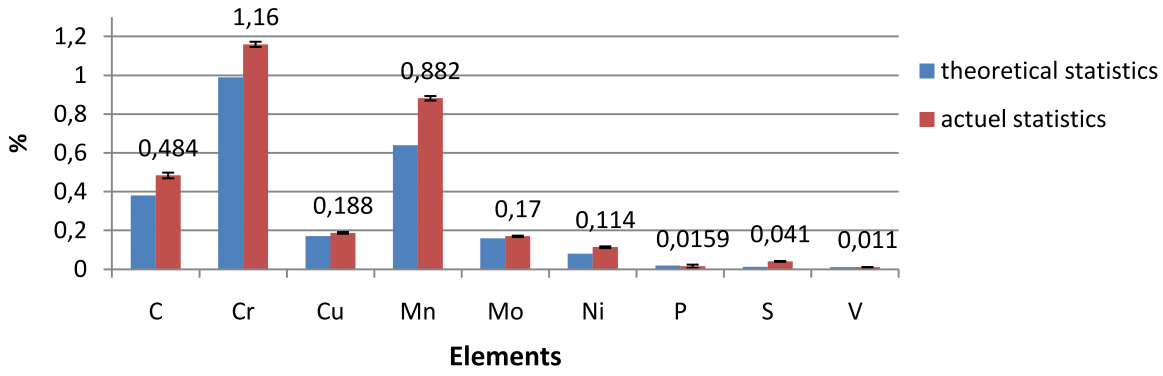


Fig.2.20. Elements analysis by spectrometry (in weight %)

Unfortunately, the elements analysis demonstrates a general excess of elements contents compared with theoretical criteria. Increasing carbon content can improve maximum hardness and strength, up to about 0.7%. When C% exceeds 0.3%, like here 0.48%, it can possibly cause cracking during quenching or welding. Relatively high S percentage leads to MnS inclusions which improves machinability.

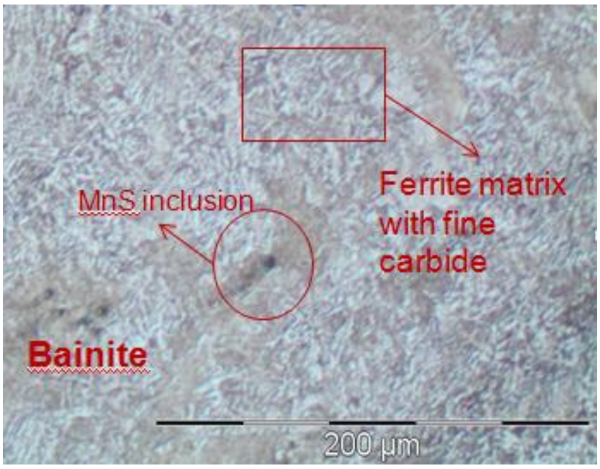


Fig.2.21. Microstructure of AISI 4140 QT(X20)

Observation of AISI 4140 QT specimen by microscope demonstrates a bainitic structure, which contains a mixture of ferrite matrix with fine carbide. The black spots represent the existing inclusion consisting of MnS. Judging from their shape; we consider that the spheroidization of this compound may have been occurred during temper. In the above figure, one can see the “tails” following the carbide caused by the polishing introduced deformation.

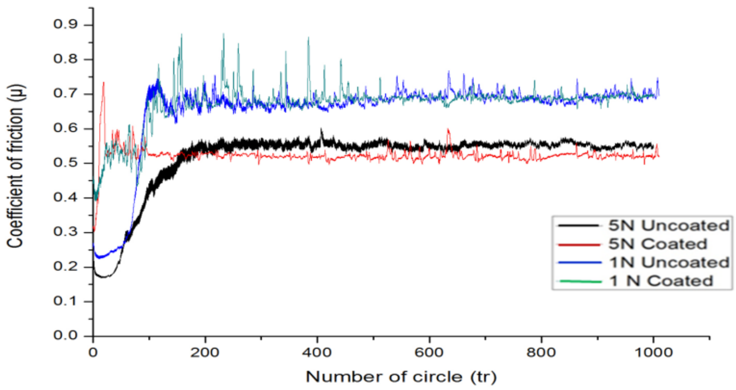


Fig.2.22 Tribology between inserts and AISI 4140 QT (results of trials done with CSM apparatus)

2.3 Design of the trials

Milling process is a machining process using a rotating tool (cutter) with usually more than one cutting edge. The primary motion is only produced by the tool and the feed motion is produced by the cutter and/or the workpiece. However, in down milling, the direction of feed of the workpiece (V_f) and the direction of rotation (V_c) of the cutting tool are the same at the point of contact ($V_c.V_f > 0$).

The down milling wear test is a delicate experience constrained by several functional specifications defining the elements of the Robin test. In this part, apparatus and experimental process are described.

2.3.1 Object & functional specifications

Our object it to obtain flank wear evolution of the reference insert and seven others treated by different edge preparations: abrasive blasting, honing, abrasive flow machining, brushing, dragging, magneto-abrasive machining, and laser machining as mentioned in chapter I. Through systematically analysis, the advantages and disadvantages of edge preparation techniques will be revealed.

Concerning the functional specifications of face down milling test, the cutting trials are carried out under the following cutting conditions:

	for AISI 304L	**for AISI 4140 QT**	**Signification**
V_c [m/min]	300	250	Cutting speed
Process	/	/	Down face milling
D_c [mm]	50	50	Tool holder diameter
f_z [mm/rev/tool]	0.36	0.485	Feed per tooth
a_p [mm]	3	3	Axial engagement
a_e [mm]	3	4.7	Radial engagement
Resulting h [mm]	0.12	0.2	Thickness of cut
Resulting l_{cu} [mm]	12	16	Length of cut
Coolant	without	without	

Tab.2.9. Cutting conditions during the trials

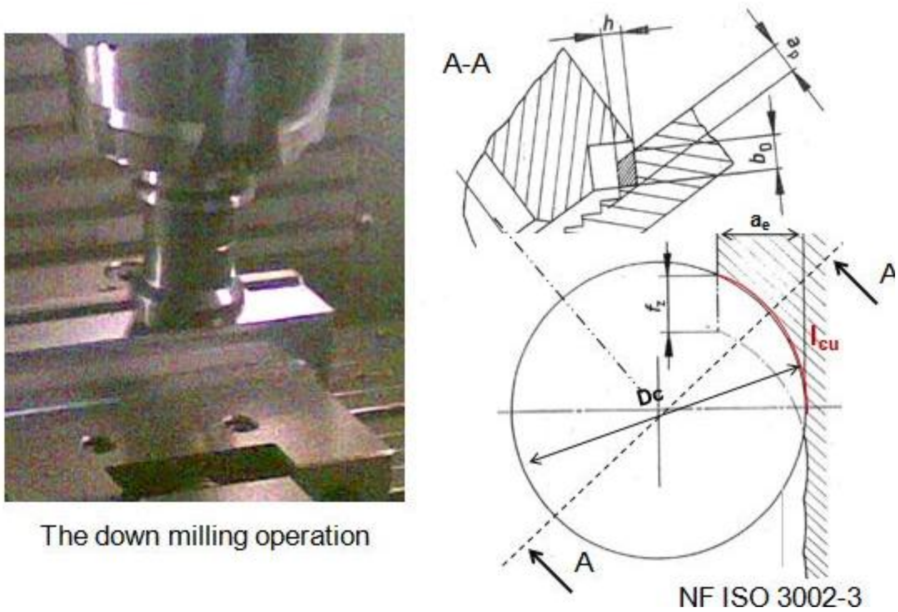


Fig.2.23. Morphology of cut during face milling [7]

It is noticeable that engagements rate is high, so, we are in the situation of a rough face milling.

2.3.2 Equipment & installation description

The object of this part is to describe apparatus installation adopted which aims to lead the trials in an appropriate way.

a) For down milling test

The down milling process requires three things: the programmed machine, the tool assembly and the workpiece assembly.

The machine used is a 3 axis NC milling machine: DMG DMC 85V with SIEMENS NC and a vertical spindle Kessler with a speed limit of 18 000 RPM and power limit of 25kW. Due to the simplicity of the tool path during milling operation, machine program is written manually in SIEMENS language and then introduced into the machine.

Fig.2.24. Tool holder, inserts used, and workpiece assembly

The workpiece assembly is composed of the primary workpiece and two clamps (fig.2.24, right)

The tool assembly is composed of the tool holder SECO R220.13-0050-12 and only one insert SECO SEAN 1203AFTN-M14. (cf. appendix 2_4 for more details)

Référence	Dc	Dc2	l1	ap*		KG		
R220.13-0050-12	50	64	48	6	4	0,6	9600	SE..1203/1303

Fig.2.25. Tool holder characteristics (extract from SECO catalogue 2012; dimensions in mm)

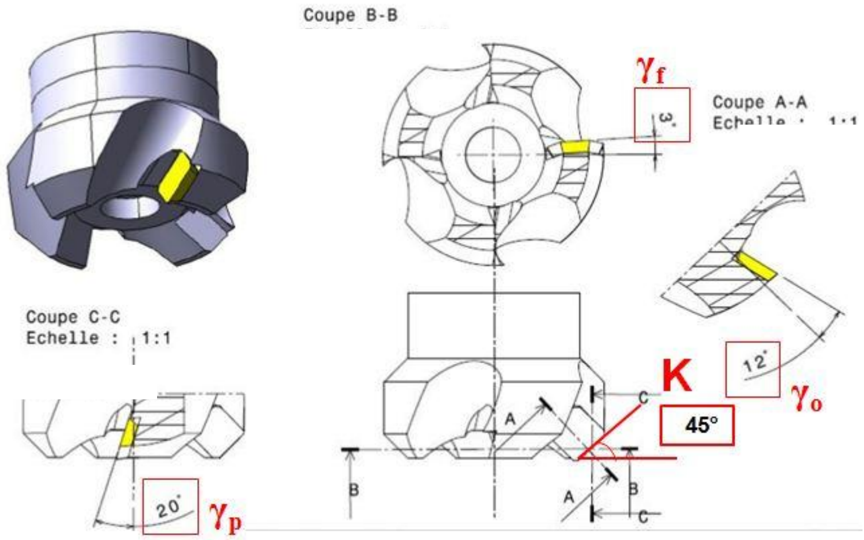


Fig.2.26. Cutting angles (according to NF ISO 3002-1 1993)

This tool has a special angle, the normal rake: γ_n as shown in the figure below:

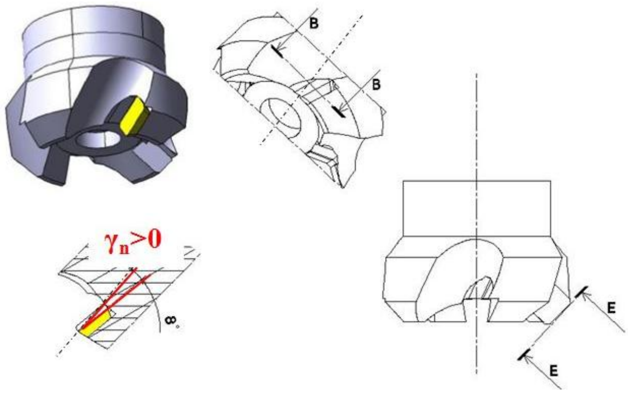


Fig.2.27. the normal rake γ_n (according to NF ISO 3002-1 1993)

This angle goes from the tool axis projection to the rake face. In our case, according to the norm ISO 3002-1, it is quantified positively. Indeed, this tool is appropriate to face mill hardly machined materials.

b) For flank wear measurement

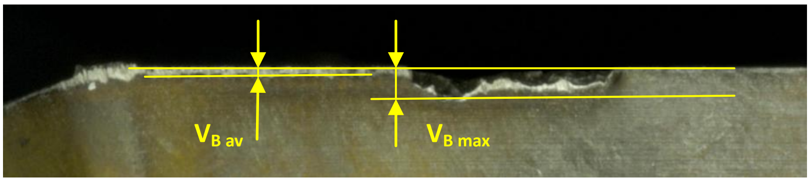


Fig.2.28. $V_{B\ max}$ and $V_{B\ av}$

Measurements concern both average flank wear $V_{B\ av}$ and maximum flank wear $V_{B\ max}$. As we see in fig.2.24, $V_{B\ av}$ describes the average wear depth of all edge area in contact with workpiece material, while $V_{B\ max}$ describes the deepest wear including spalling depth.

In the case of AISI304L milling, flank wear measurement is insured by a digital microscope with High Dynamic Range function: KEYENCE VHX-1000. In the case of 42CrMo4 QT, flank wear is measured thanks to an optical binoccular microscope Leica MZ12.

Fig.2.29. KEYENCE VHX-1000 (left) and Leica MZ12 (right)

c) For efforts measurement

For efforts measurement;

- A dynamometric transducer is used. Its role is transforming the mechanical motion into an exploitable electrical signal. It is a 9257A Kistler with the characteristics shown below:

Effort components	**Fx**	**Fy**	**Fz**
Magnitude	0-5kN	0-5kN	0-10kN
Sensibility	7.953pC/N	7.953pC/N	3.713pC/N

Tab.2.10. Kistler 9257A characteristics

- An analogic signal amplifier is used too. Its is a Kistler 5019A (4 entries), but we use only 3 entries. Its role is amplifying the electrical signals transmitted by the dynamometric transducer and providing them to the acquisition card.
- An acquisition card NI PCI M6221
- A software to visualize effort evolution during the trial, to save data and to analyse it. in our situation, it is DasyLab 11.0

The dynamometric transducer choice was constrained by the estimated efforts (cf. Appendix 2_6). In fact, the effort transducer KISTLER 9257A characteristics guarantee the possibility of measurement.

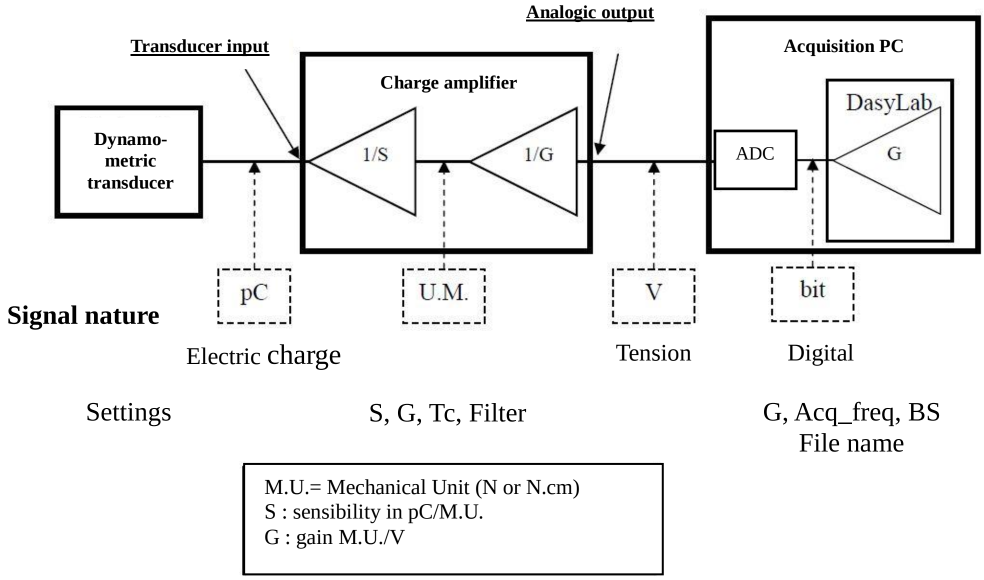


Fig.2.30 Synoptic of efforts measurement process

Before starting measurements, 4 things should be set by the associated program on DasyLab:

- **The gain G** of each one of the 3 used channels: F_x, F_y and F_z (defining measurement stretch MS and the resolution R).

In the case of measurements in F_x and F_y directions;

- Maximum estimated effort: F_x = 1050 N ; F_y = 897 N

- Measurement stretch : MS= G * (+/-10V) with |MS| > F max estimated

$\longrightarrow$ G = 200 N/V => MS = +/- 200 N/V * 10 V = +/-2000 N

- Resolution R : 16 bit acquisition card : 2^16 = 65536 levels

R = MS / 65536 (in N)

$\longrightarrow$ R = +/-2000 / 65536 = 2000 / 65536 = 0.03 N

In the case of measurements in F_z direction;

- Maximum estimated effort: F_z = 185 N

$\longrightarrow$ G = 40 N/V => MS = +/- 40 N/V * 10 V = +/-400 N

$\longrightarrow$ R = +/-400 / 65536 = 2000 / 65536 = 0.006 N

- **The acquisition frequency Acq_freq** related to the process dynamics and the pass band of the sensor. Here we are in a cutting process with variable section (the case of milling). We should have an acquisition frequency allowing at least the fundamental frequency (the frequency of the working tooth). In our current study, this frequency is calculated as below:

$f = \frac{\omega}{2\pi}$ = 32 Hz (the maximum is got in while milling AISI 304L)

Acq_freq chosen = 1000 Hz $\longrightarrow$ about 31 samples per tool revolution

- **the bloc size BS:** The recommended value is: 0.5*Acq_freq = 500
- **The file name:** Save The file as *.ddf in order to exploit it later. Every measurement should have a different file name.

2.3.3 Experimental process

A summary of the whole process used for preparation to wear measurement is presented in fig.2.31. Furthermore, appendix 2_3 contains an explanation of the whole trial protocol.

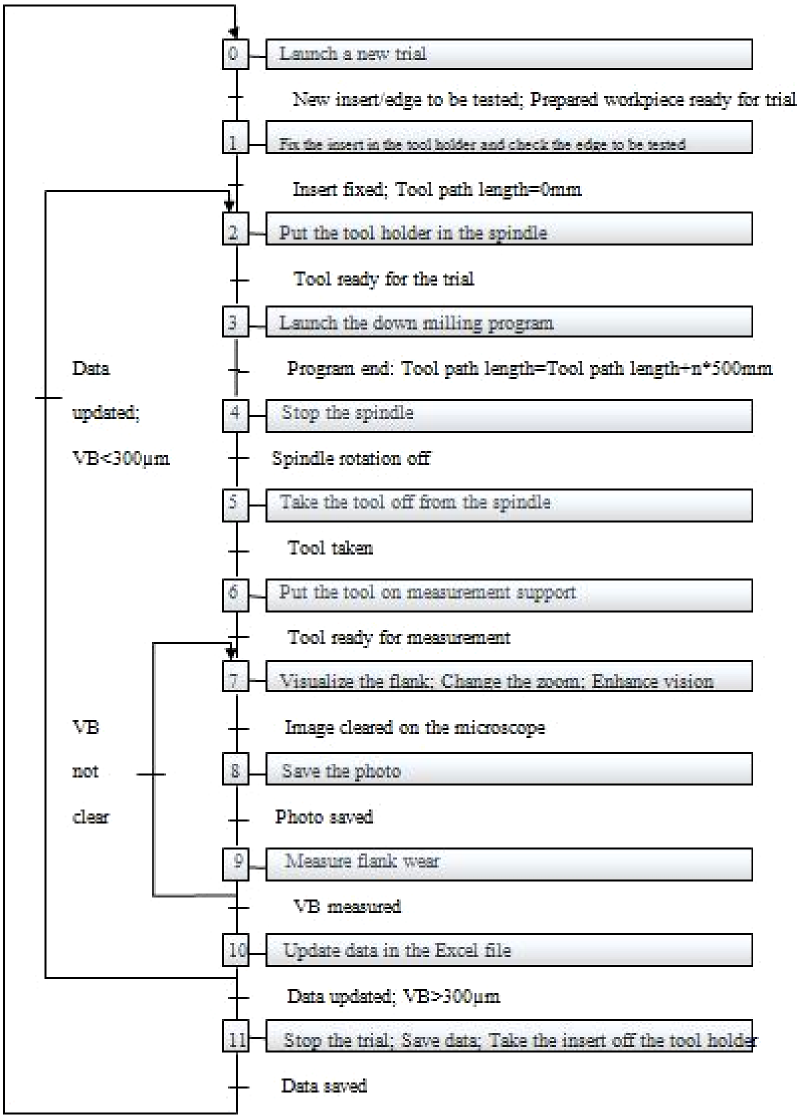


Fig.2.31. Sequential function chart of the experimental process of the current study

**n=8 for AISI304L and 5 for 42CrMo4 QT*

For effort measurement, a secondary workpiece is used. In fact, after every V_B measurement, the tool mills a workpiece attached to the dynamometric transducer. (cf. Appendix 2_5)

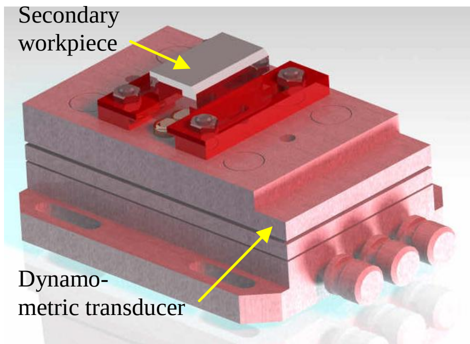


Fig.2.32. Design of the secondary workpiece assembly on the dynamometric transducer

Conclusion

In this chapter, we dealt with the study parameters concerning the insert geometry, the materials and the functional specifications ruling our down milling trials.

In the next chapter, a wear characterization will be given in order to define the relationship between the cutting edge preparation method and tool wear during down milling the stainless steel AISI 304L and the low allow steel quenched and tempered AISI 4140 QT.

Chapter 3: Wear characterization

This chapter contributes to a thoroughly qualitative analysis of tool wear including wear observation, wear mechanism explanation and influence of edge preparation on tool life for both AISI 304L and AISI 4140 QT.

3.1. Tool wear characterization during face milling AISI 304L

This part will discuss around milling on AISI 304L in the aspect of four themes: wear observation, chip analysis, complete wear chart and influence of edge preparation.

3.1.1. Wear Observation

It’s known that tool failure is a complete failure of the tool due to the destruction of its cutting edge. It’s measured on flank wear: in the zone b: V_B is quantified in µm.

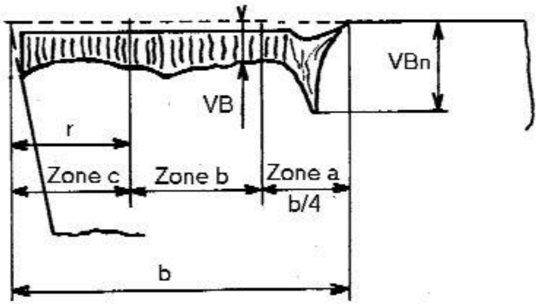


Fig.3.1. Worn tool geometry

According to ISO 3685, tool failure is "the total cutting time of a tool necessary to reach specific criterion of life". This duration is not infinite; it imposes a regular landmark to change the tool edge or the tool. So, it conditions machining time and the cost of machining a workpiece.

In our study, the specific criterion of life is reaching 300 µm of $V_{B\,max}$ for 304L (for 4140 QT, it is 200µm), and tool failure is seen as a manifestation of the phenomena below:

Name [3]	Appearance	Effects	Observed when:
Flank wear V_B		Flank wear is the most important. It has a direct impact on workpiece dimensions and the surface state. It exists on the both major and minor cutting edge	Beginning from the very beginning of all the down milling trials.
Cutting edge chipping (spallings)		Small fracture leading to a poor surface state and excessive flank wear.	Observed at the last period of tool life when approaching tool failure
Built-up edge		A chip is deposed on the insert, leading to a poor surface state	Through out the all milling process

Tab.3.1. Tool wear manifestation types seen during the trials

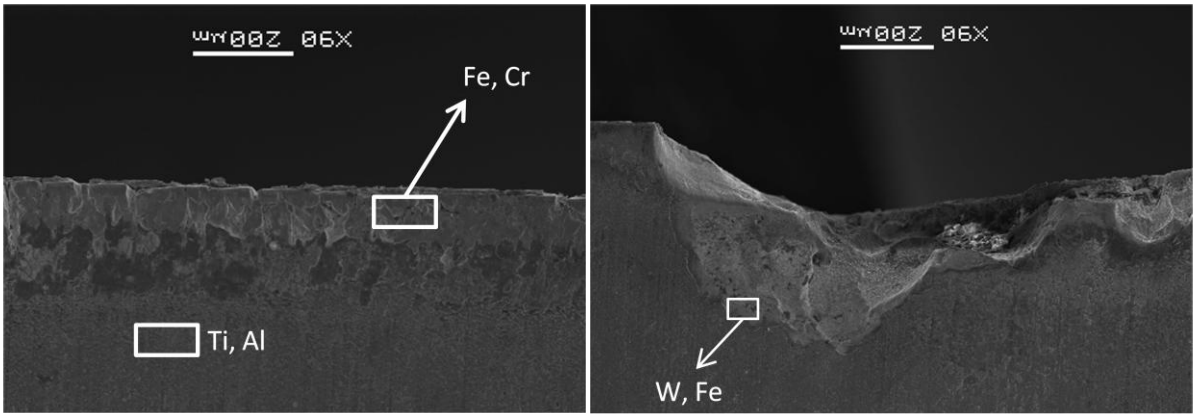


Fig.3.2. Morphology observation and EDS elements analysis of insert by SEM (major elements detected inside rectangular zones are marked)

SEM observation shows a certain content of Fe and Cr on the cutting edge surface confirming the existence of build-up edge with 304L. The right image above, the brink of spalling was detected consisting majorly of W and Fe element, which means that the big spallings appear and extend after the destruction of TiAlN layer

According to friction theory, those phenomena would be due to either adhesive wear (particles of workpiece which are deposited and cold welded to the contact surfaces of the tool and discontinuously pulled out with tool material particles) or abrasive wear (mechanical process caused by sliding and friction between the workpiece and the cutting tool).

3.1.2. Chip Analysis of AISI 304L

The roughness of the chip is closely related to the state of the cutting edge in the down milling process. Fig.3.3 is the stitching result of 304L chip. A big crude trace on the right is the trace of a big spalling on cutting edge. Ra refers to arithmetic average of absolute values, mathematically calculated as:

$$\mathrm{Ra} = \frac{1}{n}\sum_{i=1}^{n}|y_i|$$

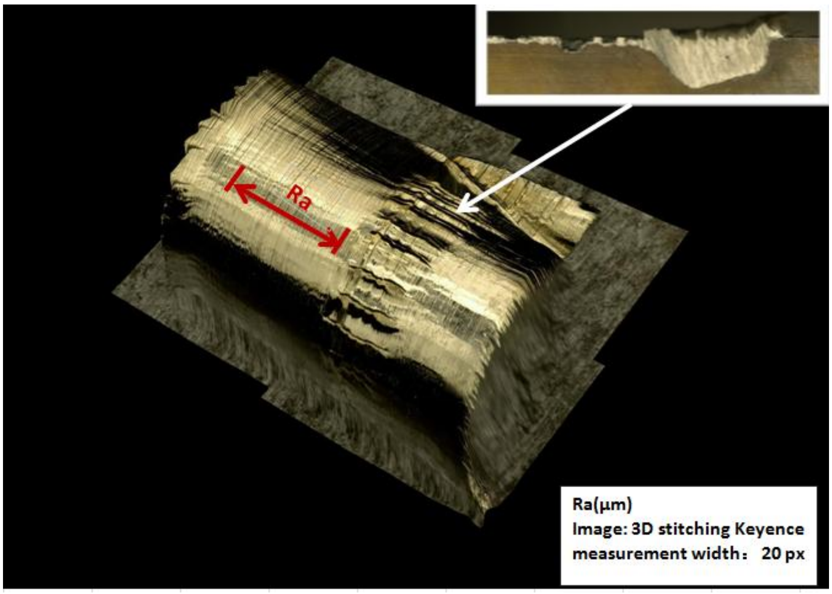


Fig.3.3 304L chip morphology by Keyence

The result shows that the roughness of the chip texture increases with the number of cuts for workpiece 304L. At the end of tool life, some chips even appeared to be broken.

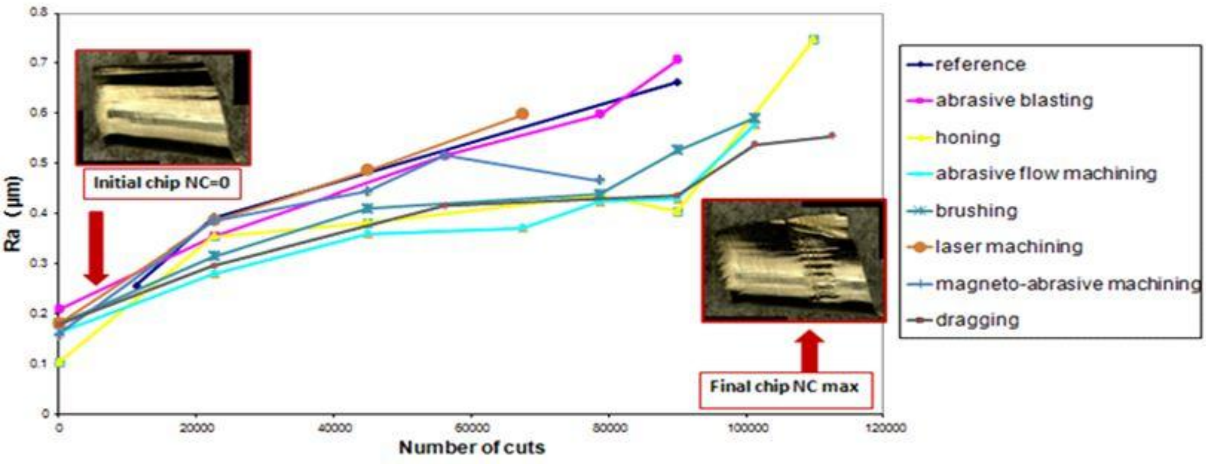


Fig.3.4 Evolution of chip roughness with number of cuts (304L)

3.1.3. Wear Charts of the inserts face milling AISI 304L

In this paragraph, the focus will be on wear charts considering $V_{B\ max}$ and $V_{B\ average}$. Since we have as many as 8 inserts involving 16 edges to exploit for 304L, here only one sample insert chart ***N°0374*** will be discussed in details, and then all the charts are gathered in two graphs: one considering $V_{B\ max}$, and the other considering $V_{B\ average}$ in order to make clear the comparison between the different preparations (cf. Appendix 3_1).

a) Sample Wear chart: the reference insert N°0374

This insert has no edge preparation. Trials were done twice using the two chosen edges: edge 3 and edge 1.

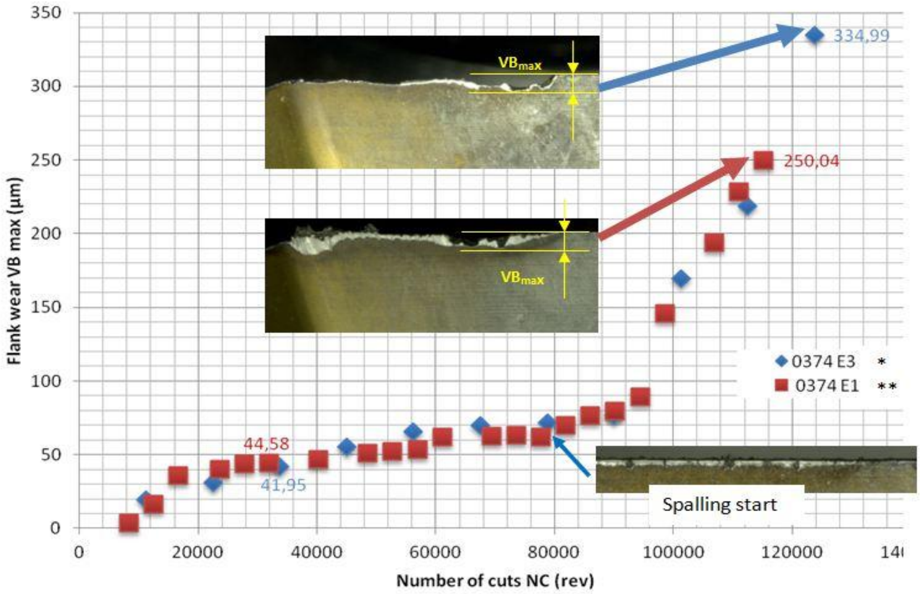


Fig.3.5. Flank wear chart of the insert 0374 down milling AISI 304L considering $V_{B\ max}$

After superposition of both flank wear charts of edge 1 and edge 3, it shows that dispersion is small and the trial is repetitive in the case of reference. The main wear mode observed is spalling in both cases, and it was the major cause of rapid wear enhancement after 90 000 rev. However, since the first trial to the last, V_B measurements were annoyed by bounded chips. They fill the spalling, and it's hard or impossible to remove them totally.

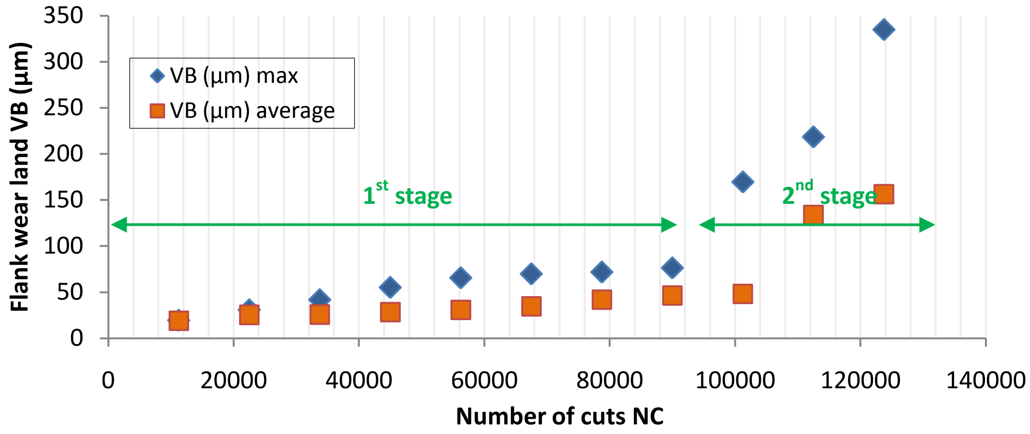


Fig.3.6. Comparison between $V_{B\,max}$ and $V_{B\,av}$ of the insert 0374

In addition, as shown in Fig.3.6, before 90 000 rev, both $V_{B\,max}$ and $V_{B\,av}$ had a linear evolution with close slopes. In fact, in this stage, micro-spalling characterizing with the maximum flank wear was not relevant, and its evolution was tardy. However, in the next stage, slope values increased too much, and a divergence between $V_{B\,av}$ rake and $V_{B\,max}$ rake was remarkable.

In the following graphs concerning $V_{B\,av}$, only the first stage will be considered. In fact, it's the period reflecting the steady evolution of flank wear.

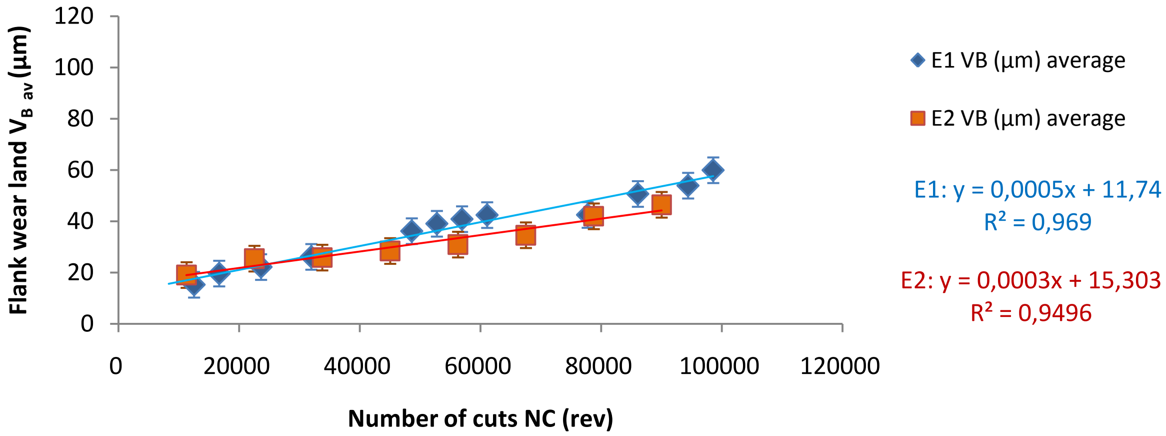


Fig.3.7. Flank wear chart of the insert 0374 down milling AISI 304L considering $V_{B\,av}$ (stage 1)

As shown in Fig.3.7, average flank wear $V_{B\,av}$ had a steady behavior, and so it could be represented by a straight line. The gap between the two edges results is small, and we conclude that the trials on the reference insert are repeatable.

b) Complete wear charts of AISI 304L

It is worthwhile to note that wear chart $V_{B\,av}$ never exceeds 300 µm (criteria for life end of the inert defined for 304L) because the appearance of huge spallings already declares the retirement of insert. In addition, observing the same range of wear from 0 to 120 m in these two charts, $V_{B\,max}$ is more dispersed. On the contrary in the chart $V_{B\,av}$, two curves of the same preparation have a smaller gap between them. However, globally, both two charts appear to be mixed up by all the curves and can be hardly given out an accurate ranking for 8 preparations. Only dragging demonstrates the best performance in tool life in either of them.

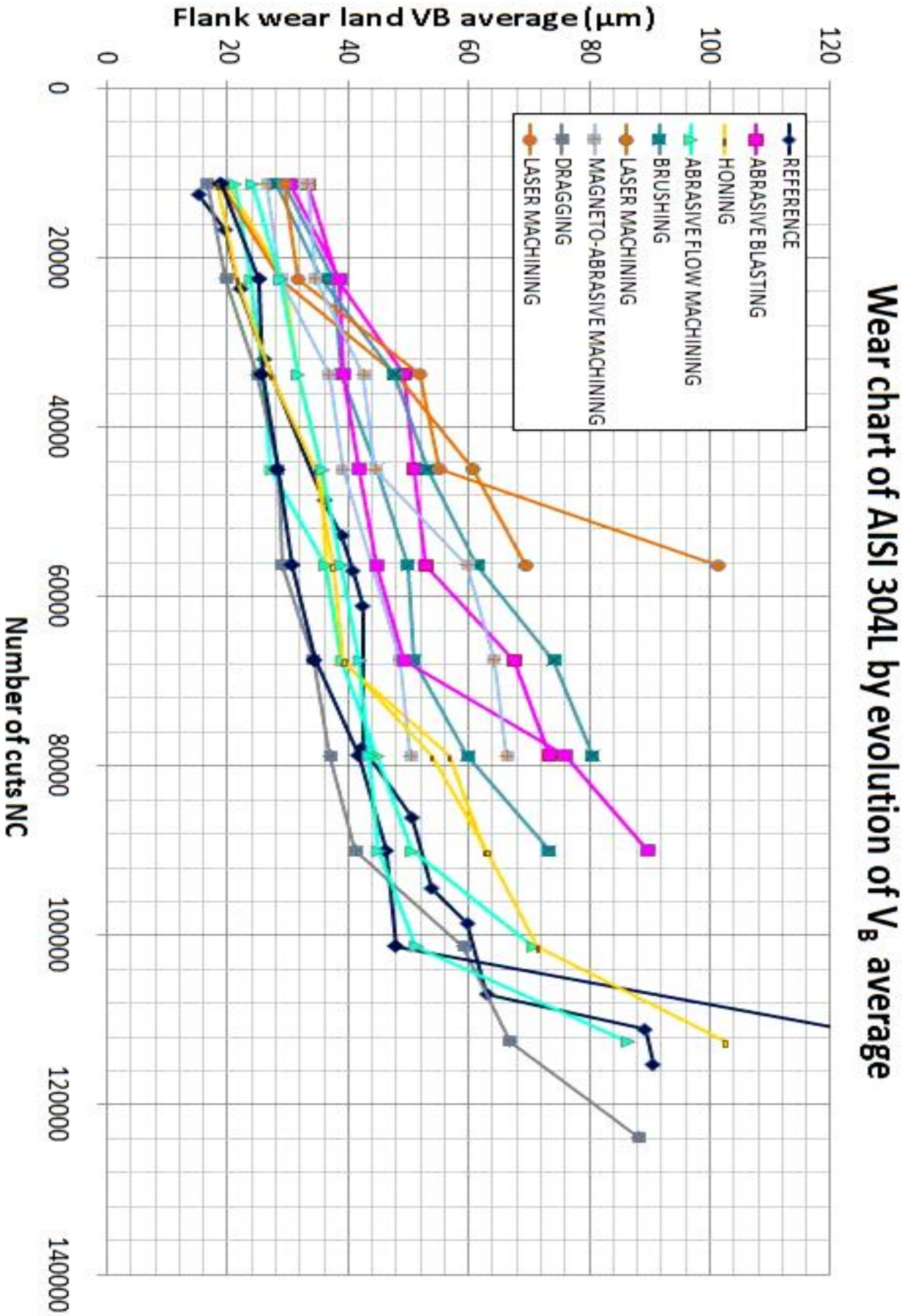


Fig 3.8 Evolution of $V_{B\,av}$(304L)

** See Appendix 3_1 for further information*

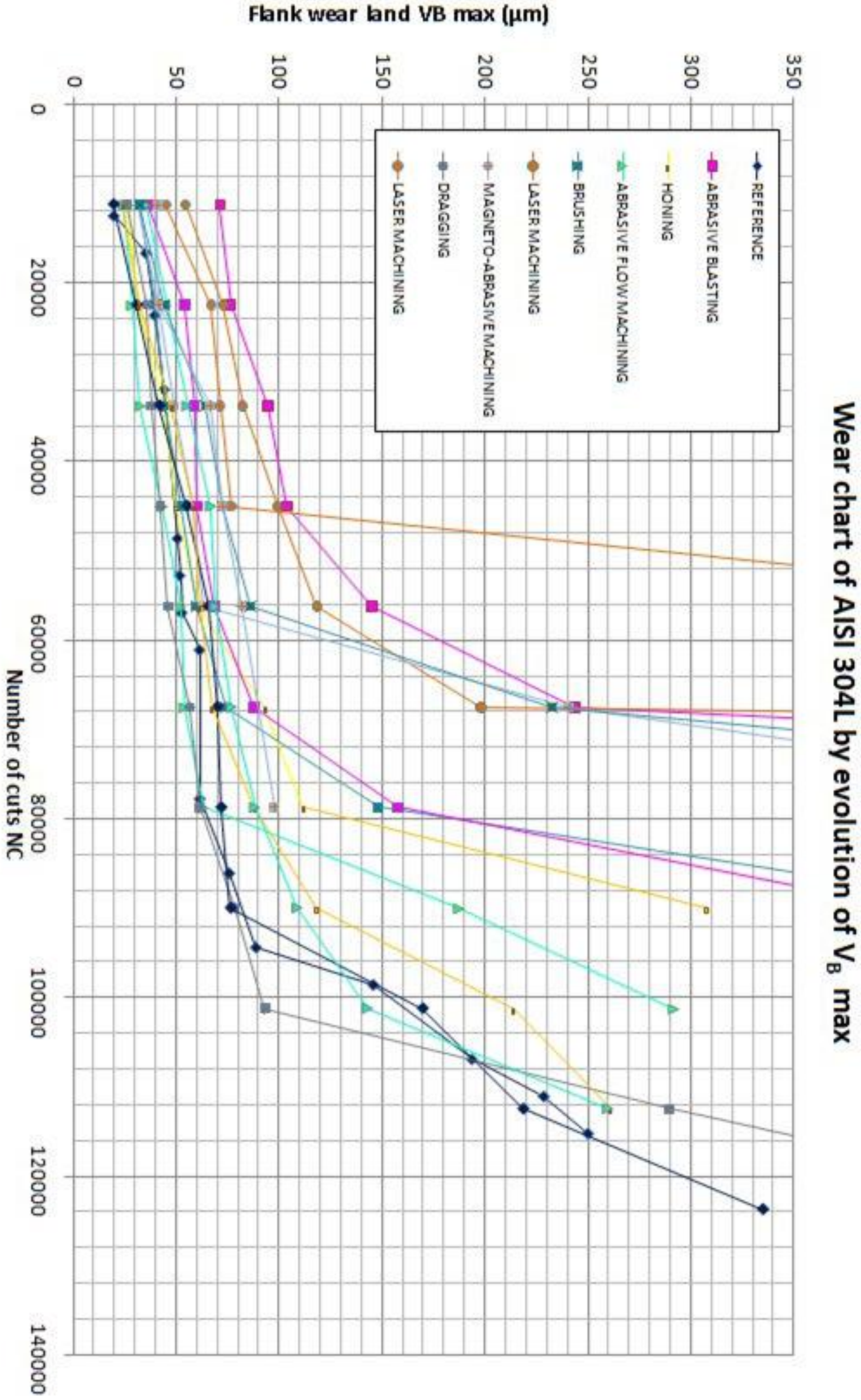


Fig.3.9. Evolution of $V_{B\,max}$* (304L)

** See Appendix 3_1 for further information*

3.1.4 Influence of edge preparation on tool life during AISI 304L milling

a) Reclassification of preparation method

According to measurements in *$ 3.1.3*, further analysis of statistical treatment of flank wear for AISI 304L has been carried out. In order to discuss the repeatability of our experiments, steady deviations (STDV) between results of two edges issued from the same insert have been calculated in terms of $V_{B\ max}$ and $V_{B\ av}$. Moreover, qualitative descriptions of spalling are also demonstrated.

	GROUP 1				GROUP 2			
Preparation	**Abrasive blasting**	**Brushing**	**Magneto-abrasive machining**	**Laser machining**	**Reference**	**Honing**	**Abrasive flow machining**	**Dragging**
STDV ($V_{B\ max}$)	312 µm	189 µm	142 µm	150 µm	81 µm	68 µm	67 µm	-
	The worst repeat-ability						The best repeat-ability	
STDV ($V_{B\ av}$)	14.32 µm	15 µm	12 µm	24 µm	34 µm	14 µm	12 µm	-
				The worst repeat-ability			The best repeat-ability	
NC when deep spalling starts (average*)	70 000	70 000	60 000	60 000	90 000	90 000	80 000	100 000
			The earliest spalling					The slowest spalling
NC when life end is reached (average*)	90 000	80 000	70 000	70 000	123 000	112 000	101 000	113 000
			The shortest tool life		The longest tool life			

Tab.3.2. Summary statement of observations

** average: average values of specified NC of 2 edges in the same insert*

Although the evaluated performances of 8 preparations method vary with the criteria that we defined (i.e. Abrasive blasting is the worst considering STDV $V_{B\ max}$ while laser machining becomes the worst taking account of STDV $V_{B\ av}$), we can classify preparations into 2 group (marked in gray) considering all information comprehensively.

Classification	**Description**	**Preparation methods**
1st group	Relatively better repeatability, slower spalling collapse, longer tool life	Reference, honing, abrasive flow machining, dragging
2nd group	Relatively worse repeatability, earlier spalling collapse, shorter tool life	Abrasive blasting, brushing, magneto-abrasive machining, laser machining

Tab.3.3. Classification of preparation

Generally, the evolutions of flank wear are classified into 3 stages: initial wear, steady wear and severe wear. The slope of the steady wear represents the rate of flank wear. We use the $V_{B\ av}$ rather than $V_{B\ max}$ to draw fig 3.10 because as we observed in tab.3.2, $V_{B\ av}$ is characterized with a smaller standard deviation which means a better repeatability and reliability.

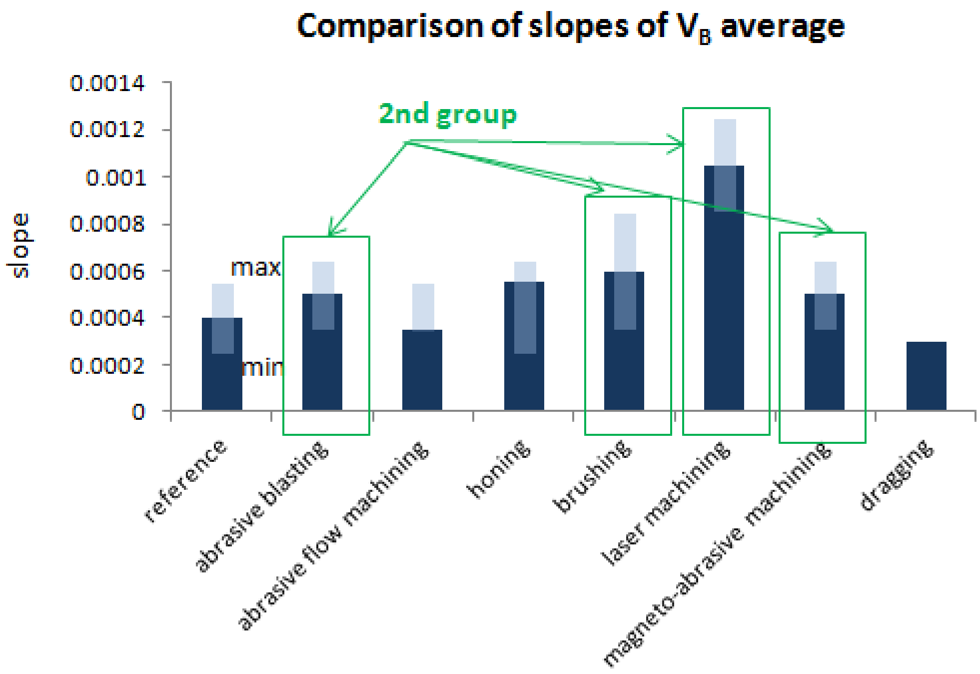


Fig 3.10. Comparison of slope values of $V_{B\ av}$ (cf. Appendix 3_2)

In fig 3.10, we made average value of the 2 slopes for every preparation. Error-bars represent variances of 2 edges compared with their average value. Despite the fluctuation, laser machining was worn fastest, while dragging turned out to be most resistant. The classification of 1st group and 2nd group is again proved as the 2nd group has a more obvious tendency of rapidly worn out.

b) Geometric influence of inserts on tool life

r_β values have already been specified by adopting the results of SurfaScan. As the major cutting edge is more engaged in the milling process, we here principally take its r_β into account instead of that of the minor cutting edge. Fig 3.11 reveals the comparison among the eight different inserts.

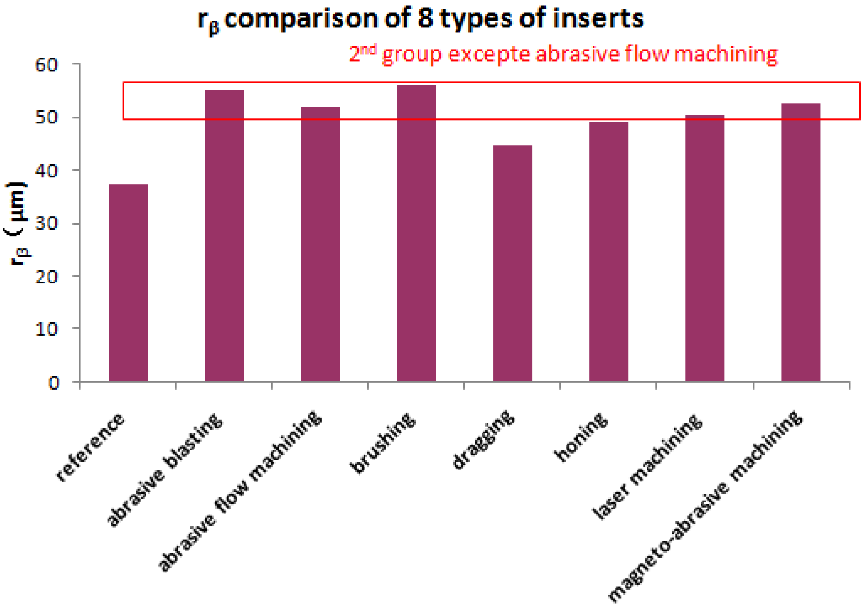


Fig 3.11. Comparison of r_β among seven preparation methods and the reference

We can observe that in 2nd group the inserts generally have a larger r_β. The only exception is the abrasive flow machining which is classified to the 1st group but it has r_β the same level as the 2nd group. In short, the preparations producing a bigger radius r_β may have a shorter tool life.

3.1.5 Conclusion about tool wear during AISI 304L milling

With the completion of all down milling tests on AISI 304L, the result of tool life can be concluded as following:

i. Preparations in the 1st group have better performance in tool life than that of the 2nd group, which means reference/honing/abrasive flow machining/dragging are better than abrasive blasting/brushing/magneto-abrasive machining/laser machining. It is failed to give a further rank or classification in either group since the differences among flank wear performance are not obvious enough.
ii. Dragging has a performance superior than the others and is characterized with a longer tool life
iii. Qualitative analysis shows a relation between insert geometry r_β and tool life, inserts with larger r_β are possibly worn out more quickly. But abrasive flow machining is an exception.

During our experiments, as we mentioned, the comparison among 8 preparations isn't very explicit. The reasons below might explicate:

i. Poor repeatability of the wearing test. During the test, the wear behavior is not ideally consistent. For example, in the test of insert 0429 edge 2, a small spalling turned up at the very beginning, in the stage of a number of cut of 11250 revolutions, which leaded to a dramatically failure of insert just after 78750 rev leaving an abnormal $V_{B\ max}$ of 1249µm. At the same time, edge 4 obtained a more reasonable curve of wear with an insert failure after 90000 rev. The big difference between the two edges might make the average V_B in Fig 3.3 unreliable.

ii. Other critical factors for tool life. In this report, we have analyzed the different factors (i.e. r_β, length of chamfer) of inserts geometry. However, the residual stress, internal stress caused by the preparation might also play a role by changing metallic phases near the cutting edge and thus, affect the tool life.

3.2. Tool wear characterization during face milling AISI 4140 QT

Another material 42CrMo4 was also used in the dry down milling. As hardened steel it has very different wear behavior. The same analysis as 304L during milling has been carried out. But only 3 cutting edges of the different 3 inserts have been tested because of lack of time.

3.2.1. Wear Observation

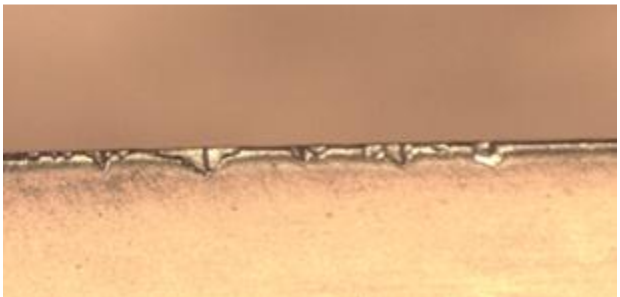


Fig.3.12. Wear image of flank face by Leica

Comparing with the former materials, 4140 QT milling leads to no phenomena of build-up edge on our tested cutting inserts. Here, tool life end is defined as exceeding 200 µm.

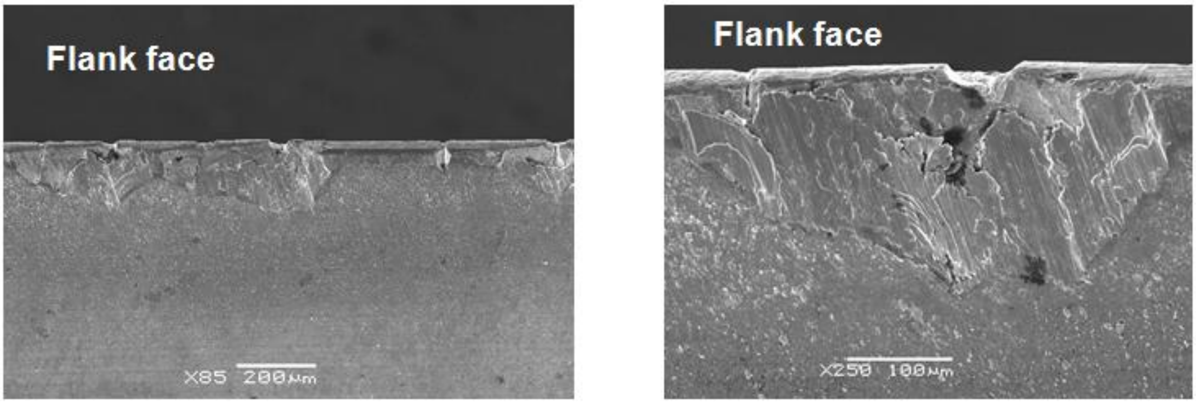


Fig.3.13. Spallings observed on the flank face by SEM

Through the above photo, the left on shows a clean cutting edge with out adhesive materials. Small crackings turned out at the beginning of tool wear. They extended and leave a spalling morphology as the right photo.

For AISI 4140 QT, the flank wear was caused by abrasive wear rather than adhesive wear.

3.2.2. Chip Analysis during 4140 QT milling

Chips shapes hardly changed from the beginning of the cutting to the tool life end in terms of curvature and roughness. Roughness remains in a range of 0.07 µm to 0.12 µm. Color changes on the chip surface caused by high temperature during milling can be seen.

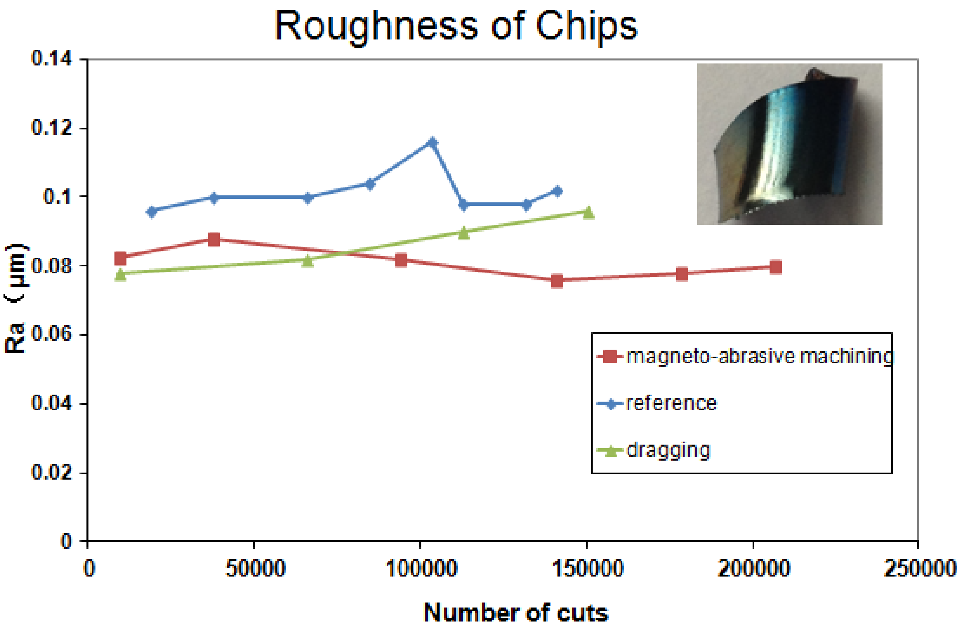


Fig.3.14. Evolution of chip roughness with number of cuts (AISI 4140 QT)

3.2.3. Wear Charts of the inserts face milling AISI 4140 QT

Because of lack of time, only three inserts ; reference, treated by dragging and treated by magneto abrasive machining were studied. On each insert one cutting edge was worn out (Fig.3.15).

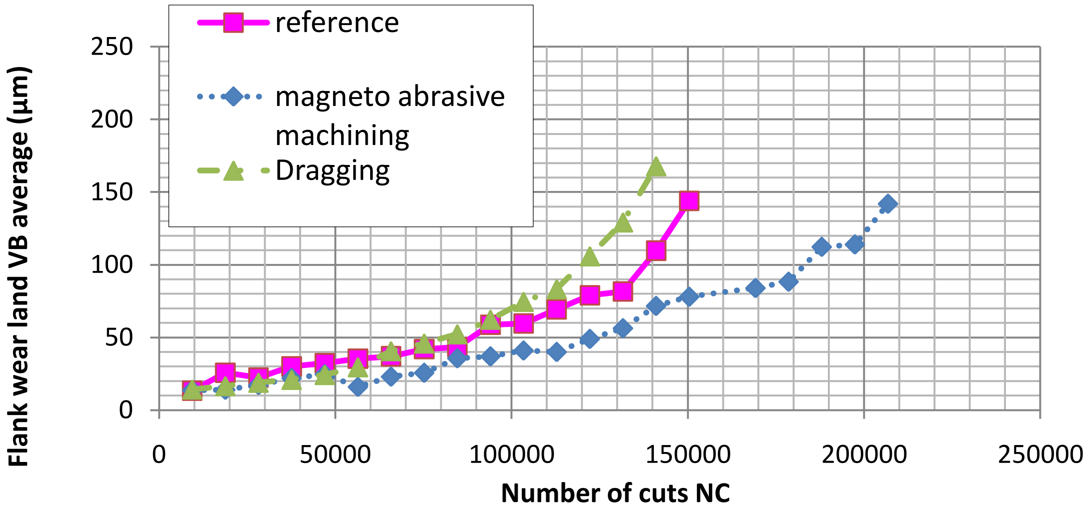


Fig.3.15.a. Evolution of $V_{B\,av}$(4140 QT)

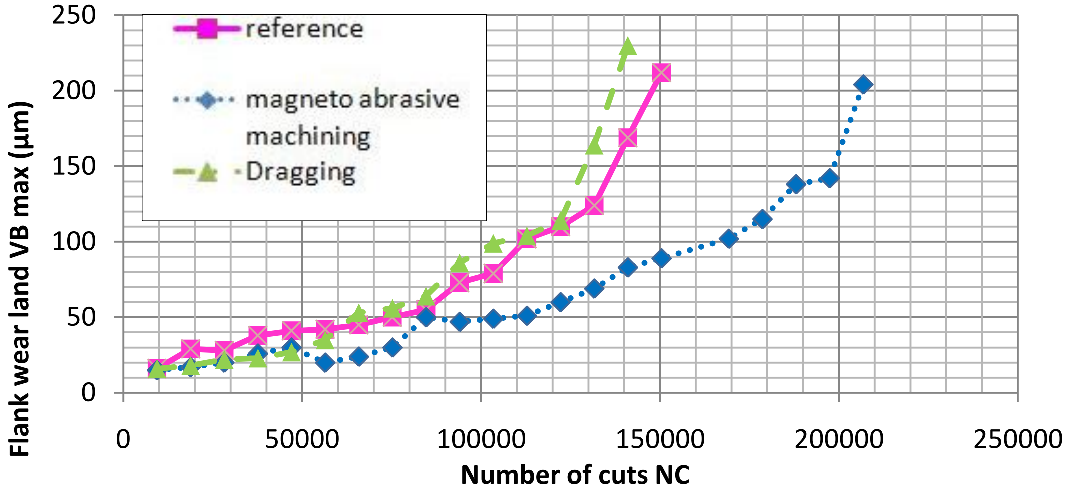


Fig.3.15.b. Evolution of $V_{B\,max}$(4140 QT)

Unfortunately, no repeating trials were done for the three inserts, but considering the inexistence of build-up edge, the repeatability of 42CrMo4 is surly better than 304L. The wear chart demonstrates that dragging has the worst wear behavior followed by reference and then magneto abrasive machining.

Conclusion

After finishing all wear milling trials on 304L and partially trials on 4140 QT, we mainly conducted a qualitative analysis on the wear results.

For 4140 QT, the preparation influences the tool life as: (from the longest to the shortest)

Magneto abrasive machining > Untreated >Dragging

It is very different to the result of milling 304L:

Untreated > Dragging > Magneto abrasive machining

To explain this discrepancy, the preparation methods give inserts different r_β , but maybe different internal stresses and other tiny transformations of the cutting edge thermo-mechanical and material state that we didn't analyze in this project. r_β contributes a part to the wear behavior but not certainly dominated. At the same time the tool wear behavior also depend on the machined materials as their properties can make a considerable impact.

Chapter 4:

Mechanical & Statistical Analysis

The object of this study is the definition of which parameter or which combination of parameters is the most characterizing wear evolution. For that, values of different variables are taken during wear trials.

Among the different statistic existing methods, principal component analysis is the most suitable to a problem treatment when it is containing measurement results. In this situation, measurement coding is relatively simple, that it consists of putting every variable in reduced centered form. This form allows mathematical handling and comparison between variables which have different units and distant magnitudes.

According to the writers of "*Méthodologie expérimentale*" [8], principal component analysis (PCA) is a mathematical technique that allows reduction of a complex correlation system into a fewer number of dimensions, in order to establish the PCA criterion. In our case, it must be correlated with flank wear land.

4.1. Mechanical analysis

4.1.1 Analysis of the workpiece efforts on the tool

Efforts measurement was done thanks to an effort transducer KISTLER 9257A and an acquisition software DasyLab 11.0. (cf. Appendix 4_1)

4.1.2 Analysis of the insert efforts on the chip

Those local efforts are issued from the measured efforts. They are then calculated by some elaborated modules on DasyLab (cf. Appendix). Those efforts are essentially Fs: The effort responsible of the shear of the chip, N, and P: the cutting force components normal and parallel to the rake face

As the insert has a working cutting edge angle κ of 45° and a working back rake γ of 20°, it was necessary to make some reference rotations in order to find the real values of those two local efforts. A number of coordinate systems is then used to resolve the cutting force into directional components. According to D.A.Stephenson & J.S.Agapiou [9], most analyses use coordinate systems in which one axis is parallel to either the cutting edge or the cutting velocity. Here cutting edge is chosen because of the oblique configuration.

4.1.3 Tool-chip friction

Friction between the tool and the chip in metal cutting influences primary deformation, built-up edge formation, cutting temperatures, and tool wear. The simplest way to characterize tool-chip friction is to define an effective friction coefficient, μ_e, as the ratio of the cutting force P parallel to the tool rake face to the force normal to the rake face N. μ_e is converted to friction angle, β, given by:

$$\boldsymbol{\beta = \arctan(\mu_e)}$$

4.1.4 Equivalent Axial Angle (EAA) and Equivalent Radial Angle (ERA)

EAA is the angle between the resultant effort and the tool revolution axis, and ERA is the angle between the resultant cutting effort and cutting velocity.

4.1.5 Shear angle

It is the angle between the cutting effort and shear plane inside the formed chip. In fact, for orthogonal cutting, Piispan, Ernst and Merchant [9] assumed that the ship is formed by shear along a single plane inclined at an angle ϕ (the shear angle) with respect to the machined surface. However, in our case, we are confronted with oblique cutting; so, it is a bit difficult to define that angle. Our reasoning was based on finding the efforts responsible of the shear Fs and defining their inclination angle relative to the resultant effort R, which we called λ.

4.1.6 Surface state parameters

Several parameters have been considered and measured on the resultant chip thanks to Wyko apparatus:

- The arithmetic mean deviation of the profile: R_a
- The maximum height of the profile: R_t
- The chip curve

4.2. Efforts and angles calculus details

As mentioned, to find local efforts, corresponding formula should be found basing on measured efforts and tool geometry.

4.2.1. Nomenclature

References	
$(\vec{e}_x,\vec{e}_y,\vec{e}_z)$	**Direct Cartesian reference (measurements reference)**
$(\vec{e}_t,\vec{e}_r,\vec{e}_z)$	**Direct cylindrical reference (considering tool revolution axis)**
$(\vec{e}_{nc},\vec{e}_r,\vec{e}_{tc})$	**Direct reference (related to cutting face)**
$(\vec{e}_{nc},\vec{e}_e,\vec{e}_{fr})$	**Direct local reference (related to cutting edge)**
Efforts	
$\vec{R}$	**Resultant effort**
$\vec{F}_x$	**Effort measured on x direction**
$\vec{F}_y$	**Effort measured on y direction**
$\vec{F}_z$	**Effort measured on z direction**
$\vec{F}_l$	**Effort projection on radial direction**
$\vec{F}_c$	**Effort projection on tangential direction parallel to cutting velocity**
$\vec{N}$	**Force normal to the rake face**
$\vec{P}$	**Cutting force parallel to the insert rake face**
$\vec{F}_p$	**Effort component parallel to the cutting edge**
$\vec{F}_{np}$	**Effort due to friction between the chip and the face**
$\vec{F}_s$	**Effort generating chip shear**
Angles	
θ_0	**Initial attack angle**
θ	**Attack angle while measurement**
κ	**Working cutting edge angle = -45°**
γ	**Working back rake = -20°**
EAA	**Equivalent axial angle, between Fz and F**
ERA	**Equivalent radial angle, between F_c and R**
β	**Mean angle of friction between the chip and the face**
λ	**Shear angle, angle between R and Fs**

Tab.4.1. Nomenclature

In order to clarify steps taken in the following calculus, a specific nomenclature is given to references, efforts and angles.

4.2.2. Base transformation matrix

The first transformation matrix M1 leads the measured efforts from the Cartesian reference, in which the transducer measures efforts, to a cylindrical reference basing on the tool revolution axis and the attack angle while measurement.

$$\mathbf{M1} = \begin{pmatrix} cos\theta & -sin\theta & 0 \\ sin\theta & cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}_{((ex\ ey\ ez) \rightarrow (et,er,ez))}$$

So, the resultant effort, in the new referential, is written as below:

$$\vec{R} = \begin{pmatrix} cos\theta & -sin\theta & 0 \\ sin\theta & cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} Fx \\ Fy \\ Fz \end{pmatrix} = \begin{pmatrix} Fx.cos\theta - Fy.sin\theta \\ Fx.sin\theta + Fy.cos\theta \\ Fz \end{pmatrix} = \begin{pmatrix} Fc \\ Fl \\ Fz \end{pmatrix}$$

The second transformation matrix M2 leads to a referential tied to the cutting face of the working insert basing on a rotation according to the working back rake γ.

$$\mathbf{M2} = \begin{pmatrix} cos\gamma & 0 & sin\gamma \\ 0 & 1 & 0 \\ -sin\gamma & 0 & cos\gamma \end{pmatrix}_{((et,er,ez) \rightarrow (enc,er,etc))}$$

So, the resultant effort is written as below in the new referential:

$$\vec{F} = \begin{pmatrix} cos\gamma & 0 & sin\gamma \\ 0 & 1 & 0 \\ -sin\gamma & 0 & cos\gamma \end{pmatrix} \begin{pmatrix} Fc \\ Fl \\ Fz \end{pmatrix} = \begin{pmatrix} Fc.cos\gamma + Fz.sin\gamma \\ Fl \\ -Fc.sin\gamma + Fz.cos\gamma \end{pmatrix} = \begin{pmatrix} N \\ Fl \\ Ftc \end{pmatrix}$$

The third and last transformation matrix M3 leads to a referential that considers the major cutting edge as a principle direction, basing on a rotation according to the working cutting edge angle K. Here, the plan ($\vec{e}_{nc}$,$\vec{e}_{e}$) is the shear plan of the chip.

$$\mathbf{M3} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & cos\kappa & -sin\kappa \\ 0 & sin\kappa & cos\kappa \end{pmatrix}_{((enc,er,etc) \rightarrow (enc,ee,efr))}$$

Finally, the resultant effort can be written in the local referential as below:

$$\vec{F} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & cos\kappa & -sin\kappa \\ 0 & sin\kappa & cos\kappa \end{pmatrix} \begin{pmatrix} N \\ Fl \\ Ftc \end{pmatrix} = \begin{pmatrix} N \\ Fr.cos\kappa - Ftc.sin\kappa \\ Fr.sin\kappa + Ftc.cos\kappa \end{pmatrix} = \begin{pmatrix} N \\ Fnp \\ Fp \end{pmatrix}$$

$$\vec{F} = \begin{pmatrix} (Fx.cos\theta - Fy.sin\theta).cos\gamma + Fz.sin\gamma \\ (Fx.sin\theta + Fy.cos\theta).cos\kappa - (-(Fx.cos\theta - Fy.sin\theta).sin\gamma + Fz.cos\gamma).sin\kappa \\ (Fx.sin\theta + Fy.cos\theta).sin\kappa + (-(Fx.cos\theta - Fy.sin\theta).sin\gamma + Fz.cos\gamma).cos\kappa \end{pmatrix} = \begin{pmatrix} N \\ Fnp \\ Fp \end{pmatrix}$$

4.2.3. Formula

Here all formula needed are collected.

$\vec{R} = \vec{F}_x + \vec{F}_y + \vec{F}_z$	$\vec{R} = \vec{N} + \vec{P}$	$\beta = (\vec{N} \,\hat{;}\, \vec{R})$	β = arctan (P/N)
$\vec{P} = \vec{F}_p + \vec{P}_n$	$\vec{F}_s = \vec{N} + \vec{F}_p$	ERA = $(\vec{R} \,\hat{;}\, \vec{F}_c)$	ERA = arccos(F_c/R)

EAA = $(\vec{Fz} \hat{,} \vec{R})$ EAA = arcos(F_z/R) $\lambda = (\vec{R} \hat{,} \vec{F}_s)$ λ= arcos(Fs/R)

4.2.4. Modeling

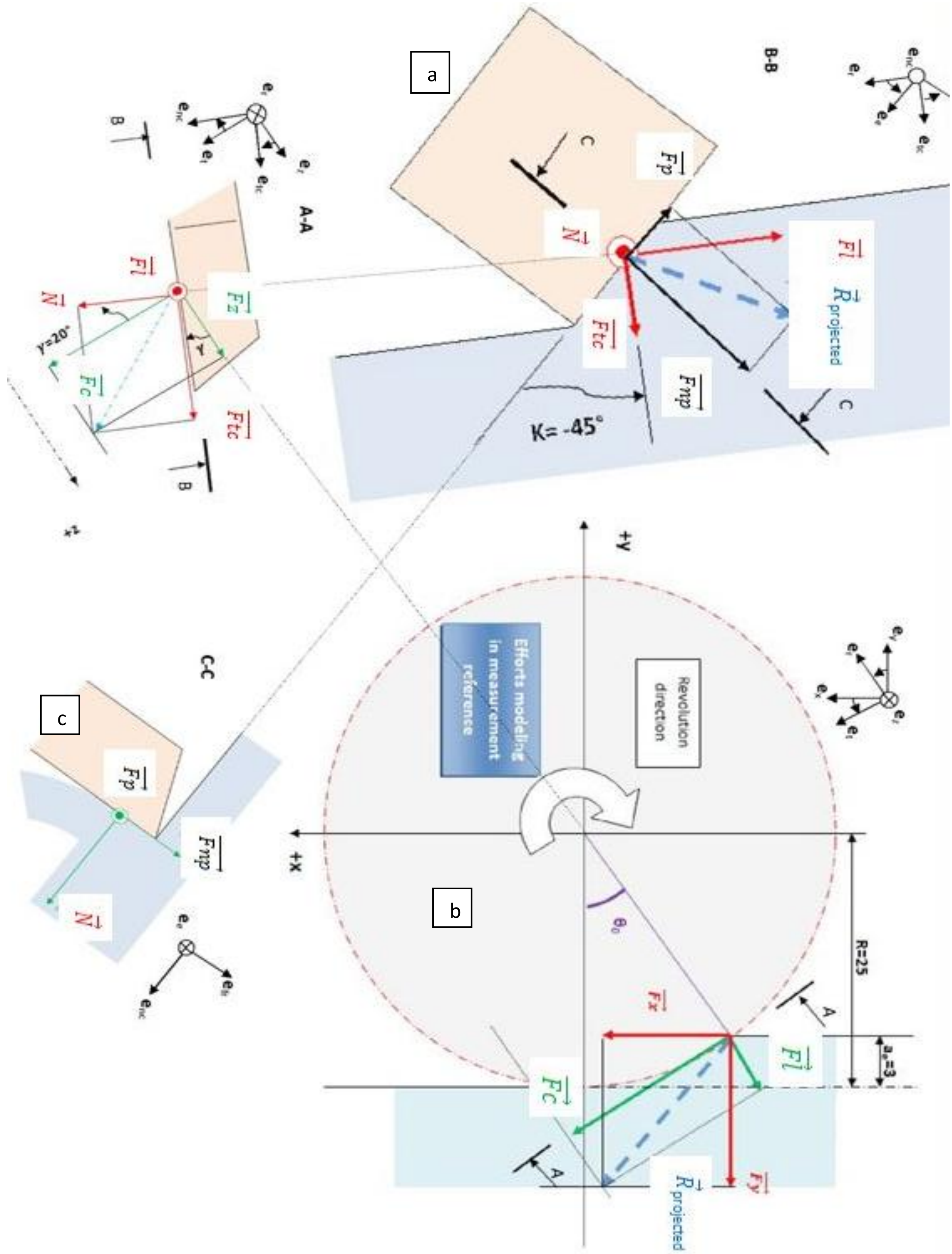


Fig.4.1. Efforts modeling (2D drafts)

The fig.4.1 explains cutting effort components used in the previous calculus, their position in the global and local references, and between the insert and the chip.

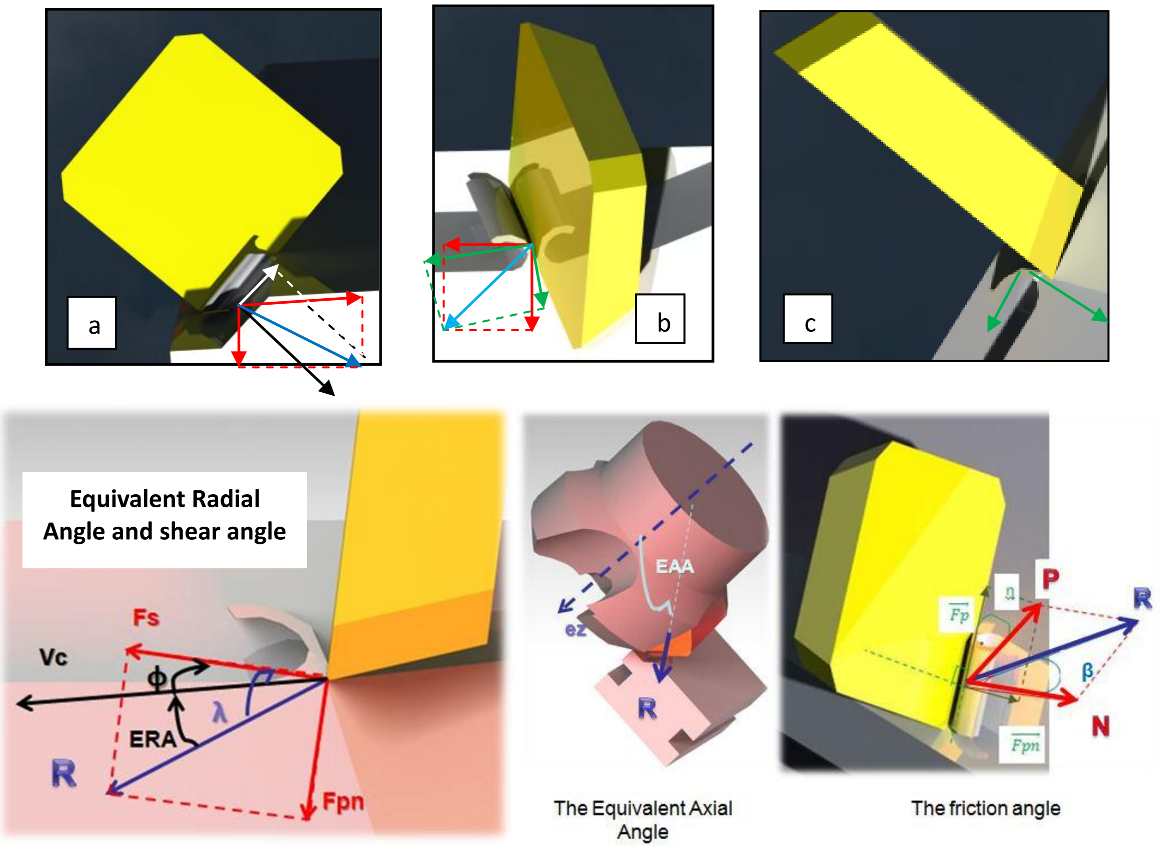


Fig.4.2. Local efforts and angles modeling (3D drafts)

4.2.5. Local entities evolution during the face milling trials: ERA evolution as an example

The local entities evolution during trials is surveyed thanks to modules developed on DasyLab basing on the previous calculus. However, the transducer gives information continuous in time, while the cutting tooth attacks the work material in periodic cycles. So, we should get rid of useless data afforded. In order to clarify this handling, in this paragraph, we study the example of the equivalent radial angle evolution:

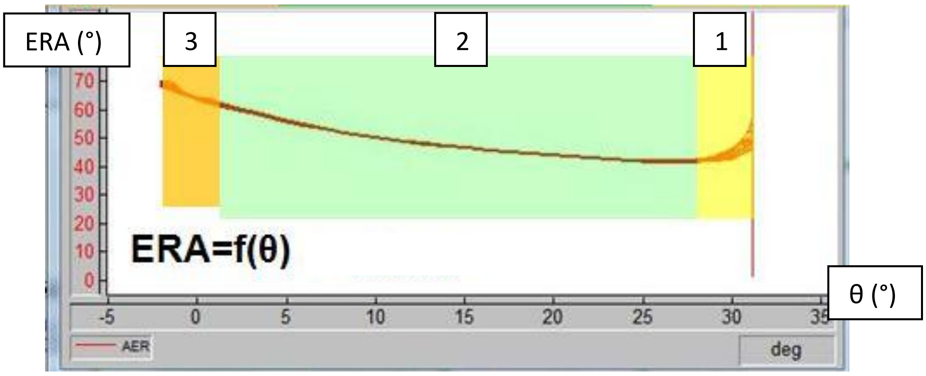


Fig.4.3. ERA evolution with attack angle θ by DasyLab

Fig.4.3 shows the evolution of the equivalent radial angle according to the attack angle θ position (the red curve). Three could be distinguished:

- Zone 1: Unstable, decreasing evolution
- Zone 2: Stable, slightly increasing
- Zone 3: Increasing, with higher rake.

Although we have only conserved curves while the tooth works on the material, instability of the signals is seen at the beginning and the end of the plots. This has an explanation.

In fact, the contact between the cutting edge and the work material knows three regimes:

- The first one, manifested in zone 1, corresponds to the partial engagement of the cutting edge in the

workpiece, and the attack angle goes from θ_0 to θ_1.
- The second one, seen in zone 2, is a permanent regime. It corresponds to the complete engagement of the cutting edge in the work material, and the attack angle goes from θ_1 to θ_2.
- The third and last one, zone 3, is seen when the tooth is partially getting out, and the contact interface is diminishing, and the attack angle goes from θ_2 to θ_3.

However, lengths of the two transitory zones are measured according to the axial and radial engagements: a_e and a_p, that is why the treatment program on DasyLab while dealing with 304L is different from that of 4140 QT.

The next figure gives a geometric explanation of those regimes, considering ERA and θ evolution.

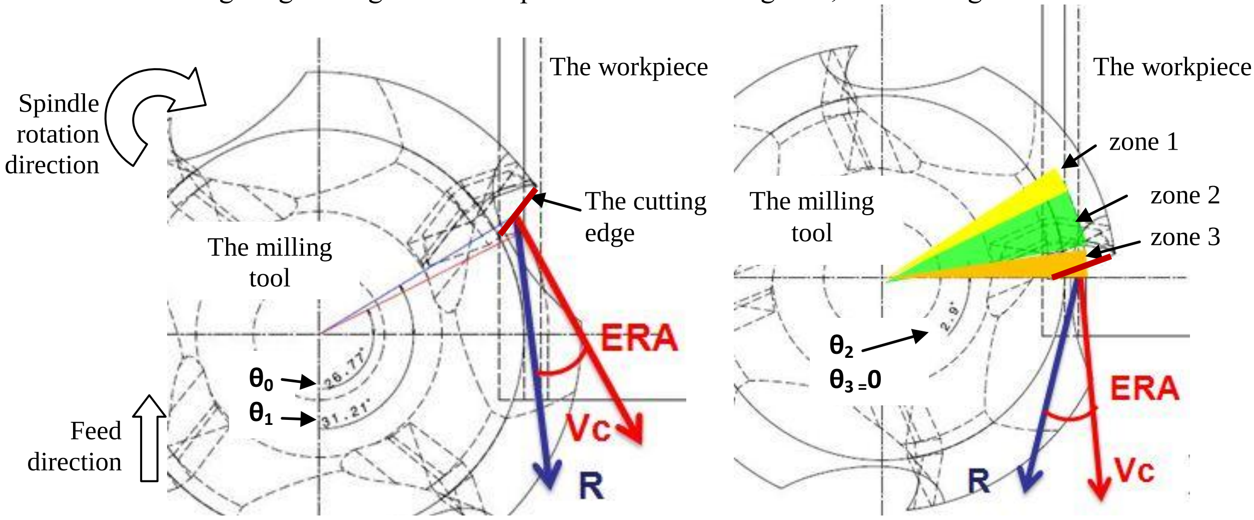


Fig.4.4. ERA evolution with the attack angle during AISI 304L milling

In order to guarantee the efficiency of the measurements done on DasyLab, we keep only values calculated on the permanent regime, and we pick up the average and maximum ones.

4.3. Principal Component Analysis

The next graphic represents the evolution of 1 among the 21 variables measured and calculated during 14 wear trials done on AISI 304L. This shows the problem of results exploitation without mathematical analysis. We should notice that we are looking for variables curves behaving like V_B.

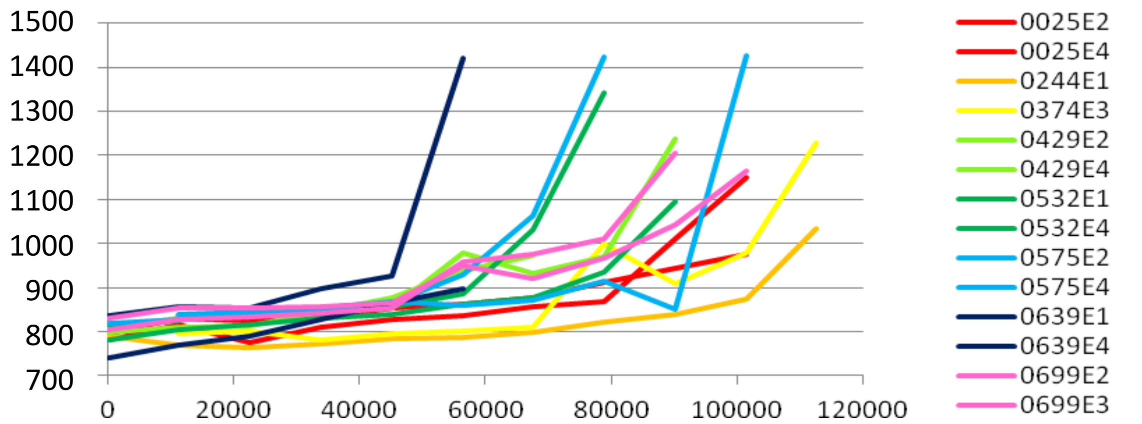


Fig.4.5. Maximum resultant effort evolution: R max [N] =f(NC[rev])

As we see in fig.4.5, it is possible to have an idea about criteria following the evolution of wear. However, the definition of the parameter the most reflecting wear needs a mathematical method. A principal component analysis method applied to our trials results is then necessary. For that, values of the different parameters in addition to the associated V_B must be written in a table.

The next organization chart presents the principle of the principal component analysis (PCA) method explained by C.Blanc and R.Marotte [10]. As we have a high number of surveyed variables (21 for both AISI 304L and 42CrMo4), and the complete number of wear trials, it is impossible to report the found results step by step.

Principle of the PCA applied to a simple sample

	Var1	**Var2**	**Var3**
Trial1			
Trial2			

	Var1	**Var2**	**Var3**
Trial1			
Trial2			

= (value-mean value of Var1)/steady deviation(Var1)

	Var1	**Var2**	**Var3**
Var1	1		
Var2		1	
Var3			1

= correlation coefficient between Var1 and Var2

Two parameters are highly correlated (correlation coefficient close to 1) present a redundancy in information: We can get rid of one of them.

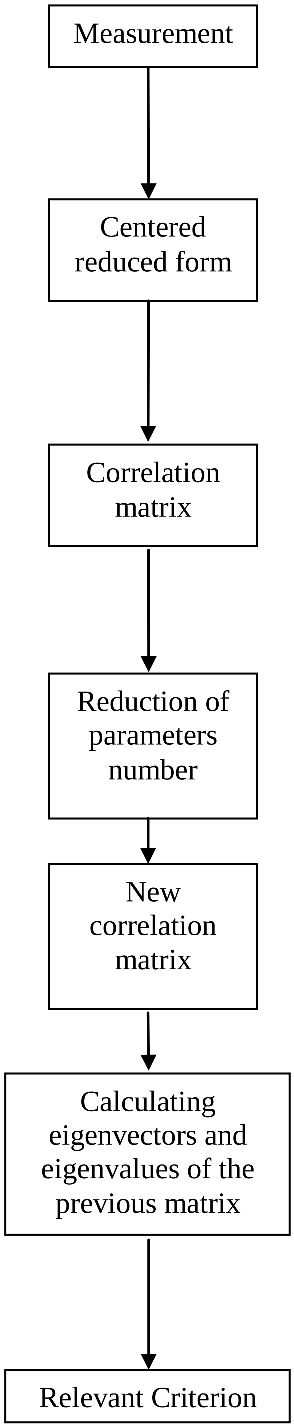
	Var1	**Var2**
Var1	1	
Var2		1

	EigVal1	**EigVal2**
	x	y

	EigVect1	**EigVect2**
Var1	a	c
Var2	b	d

Eigenvalues show the quantity of data included in the associated eigenvector. If x>y, the vector EigVect1 is the relevant criterion.

$$\mathbf{RC=a*var1+b*var2}$$



4.3.1. PCA applied to AISI 304L results

a) Centered reduced form

This form is written basing on the mean value and the steady deviation of a serial measurement x_i. The new variable X_i becomes then without unit.

$$X_i = \frac{xi - \bar{x}}{\sigma x}$$

The use of this form allows the summation of entities with different physical significations without losing sense. In fact, a table is elaborated regrouping comparable values (cf Appendix 4_2).

b) Correlations analysis

Basing on values in centered reduced form, linear correlation coefficients are calculated. Results are gathered in a symmetric square matrix called correlation matrix. (cf. Appendix 4_2)

Here, only variables having a correlation coefficient with $V_{B\ max}$ exceeding 70% are conserved, that we took this scale as a limit in the repartition of the coefficients.

So, only 6 parameters are left to represent wear evolution: F_p, F_s, P, F_z ,R and EAA. However, as they are highly correlated between themselves, only parameters remaining to the same referential are left, which are: $F_{z\ max}$, R_{max}, and EAA_{max}.

Now parameters implication percentage should be defined. For that, eigenvalues and eigenvectors of the simplified matrix should be calculated.

	$F_{z\ max}$	R_{max}	EAA_{max}
$F_{z\ max}$	1,0000	0,9345	-0,8675
R_{max}	0,9345	1,0000	-0,6974
EAA_{max}	-0,8675	-0,6974	1,0000

Tab.4.2. Simplified Correlation matrix for AISI 404L

c) Eigen values

The analysis of the linear correlation matrix allows calculus of its eigenvectors and eigenvalues. Eigenvectors are called factors. Eigenvalues are reflecting the amount of information included in every factor. After ranking them, we are able to analyze these results.

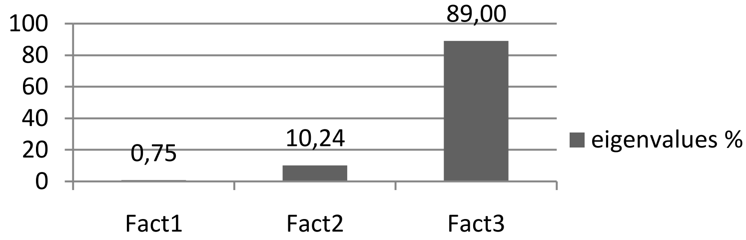


Fig.4.6. Eigenvalues implication percentage (for AISI304L)

As we see, the first factor of its own represents 89% of the information. It is so the relevant criterion.

	Fact1	Fact2	Fact3
$F_{z\ max}$	0,7887	0,0988	-0,6067
R_{max}	-0,5187	0,6367	-0,5706
EAA_{max}	0,3300	0,7648	0,5534

Tab.4.3. The different factors (for AISI 304.L)

d) The relevant criterion

According to the previous paragraph, we choose factor 1 as a relevant criterion RC. So,

$$RC= -0.607*F_{z\ max} -0.571*R_{max} +0.\ 553*EAA_{max}$$

In order to give a physical signification to the coefficients of the relevant criterion, we will come back to a percentage of the whole information. i.e: With a coefficient of -0.6067, $F_{z\ max}$ represents

97.24% of Fact3, R_{max} 91.46%, and EAA_{max} -88.70%. (cf. Tab.5.3.)

	Fact3 bis
$F_{z\,max}$	97.24%
R_{max}	91.46%
EAA_{max}	-88.70%

Tab.4.4.Percentage of the implication of every variable in Fact3

However, factor 3 bis, like factor 3, represents 89% of the information as it is associated to the third eigenvalue (cf. Fig.4.6). By multiplying every value of the factor 3 bis by 0.89, we will get the percentage of the total information included in every variable. (cf. tab.5.4)

	% total info
$F_{z\,max}$	86,55%
R_{max}	81,40%
EAA_{max}	-78,95%

Tab.4.5.Percentage of the implication of every variable in the total information

Now the expression of the RC becomes:

$$\mathbf{RC= -0.865*F_{z\,max} -0.814*R_{max} +0.\ 789*EAA_{max}}$$

Let's check that maximum flank wear can effectively be characterized by this criterion. For that, we will just see the correlation coefficient relating RC to $V_{B\,max}$ corresponding to each preparation:

Preparation method	Brushing	Laser machining	Abrasive flow machining	Dragging	Magneto abrasive machining	Honing	Abrasive blasting	Reference
Correlation %	99.16	98.92	97.34	97.94	96.70	96.04	94.71	86.63

Tab.4.6. Correlation % between the RC and $V_{B\,max}$ (for AISI304L)

As we see in the previous figure, RC characterizes $V_{B\,max}$ in a reliable way.

e) From the relevant criterion to wear law

As the RC is highly correlated with $V_{B\,max}$, we tried to find the affine relation between them for each edge preparation. The statistical treatment led to the results below:

Preparation	Mathematical relation between RC and VB max
Abrasive flow machining	VB max = f(RC) = 0,0753.RC - 0,1906
Dragging	VB max = f(RC) = 0,1071.RC - 0,0719
Reference	VB max = f(RC) = 0,0955.RC - 0,0438
Abrasive blasting	VB max = f(RC) = 0,1129.RC - 0,1114
Magneto abrasive machining	VB max = f(RC) = 0,1281.RC - 0,1634
Brushing	VB max = f(RC) = 0,1345.RC - 0,1744
Laser machining	VB max = f(RC) = 0,1116.RC - 0,1487
Honing	VB max = f(RC) = 0,0759.RC - 0,1919

Tab.4.7. Mathematical relation between the RC and $V_{B\,max}$

Basing on the equations found, $V_{B\,max}$ evolution graphs obtained by the RC are compared with graphs obtained by measurements. In this case, values are kept in reduced notation (unitless). The graphs shown below are examples of the results found, the other graphs are given in appendix.

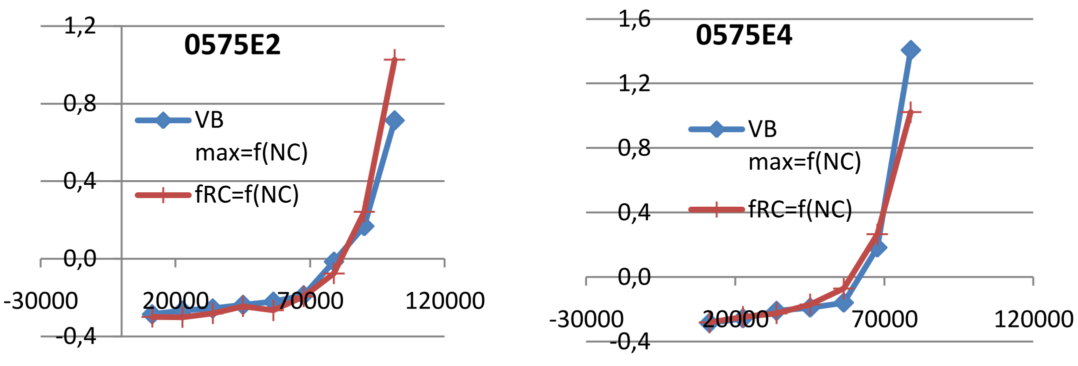


Fig.4.7. $V_{B\,max}$ evolution ($V_{B\,max}$=f(NC [rev])) obtained by measurement (VB max)) and by RC (fRC) in case of brushing (edge 2 on the left, edge 4 on the right)

As we see in the previous figure (fig.4.7), in case of brushing, curves collapse. However, it is not the case of all preparations. Although the correlation is high for all cases, the error margin is still high especially for the reference. (cf. appendix 4.2)

4.3.2. PCA applied to AISI 4140 QT results

Concerning the hardened low alloy steel AISI 4140 QT, we led trials on three inserts: the reference, the insert prepared by dragging, and the insert prepared by magneto-abrasive machining. 21 variables were calculated basing on results acquired by the transducer (cf. Appendix 4_3).

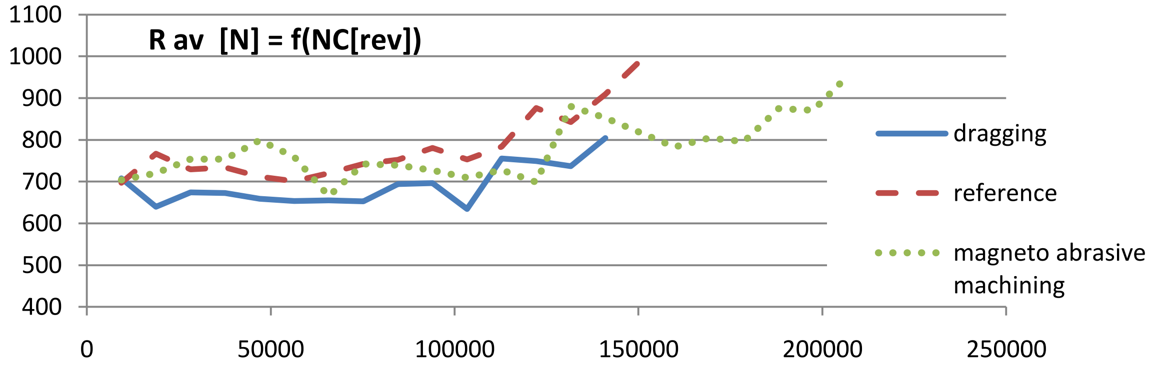


Fig.4.8. Resultant effort R evolution with the number of cuts NC

As shown in fig.4.8, the resultant effort mostly increases with the number of cuts. However, there is no evidence concerning its correlation with $V_{B\,max}$. So, R and the twenty other parameters were subject to the PCA in order to get a coherent relation between efforts and wear.

a) Correlations analysis

In this analysis, only variables having a correlation coefficient with $V_{B\,max}$ exceeding 62% are conserved, that we took this scale as a limit in the repartition of the coefficients. Comparing with AISI 304L, this limit seems lower. In fact, 4140 QT results have shown poor correlation between $V_{B\,max}$ and the other variables. So we were obliged to lower it. (cf. Appendix 4_3 for correlation matrix)

In this case, only 3 parameters are left to represent wear evolution: $F_{p\,av}$, P_{av}, $F_{z\,av}$ and R_{av}.

	$F_{p\ av}$	$F_{pn\ av}$	R_{av}
$F_{p\ av}$	1,0000	-0,7483	0,9907
$F_{pn\ av}$	-0,7483	1,0000	-0,7504
R_{av}	0,9907	-0,7504	1,0000

Tab.4.8. Simplified Correlation matrix for 4140 QT

b) Eigen values

After a ranking of the factors calculated basing on the simplified correlation matrix, eigen values allowed us to classify the factors according to their implication percentage in the collected data.

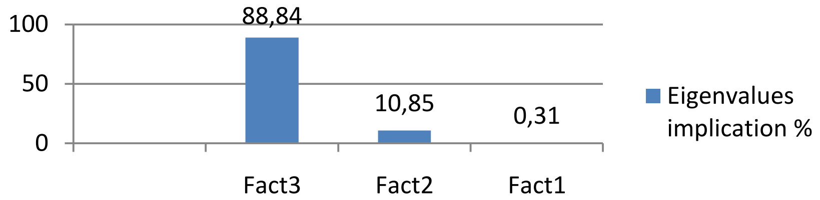


Fig.4.9. Eigenvalues implication percentage (for 4140 QT)

As shown in fig.4.9, factor 3 represents 88.84% of the information. It is the relevant criterion.

	Fact1	Fact2	Fact3
$F_{p\ av}$	-0,7058	0,3825	0,5963
$F_{pn\ av}$	0,0035	0,8436	-0,5369
R_{av}	0,7084	0,3769	0,5968

Tab.4.9. The different factors (for 4140 QT)

c) The relevant criterion

According to the previous paragraph, we choose factor 3 as a relevant criterion RC. So,

$$RC= 0.596*F_{p\ av} -0. 537*F_{pn\ av} +0.597*R_{av}$$

In order to give a physical signification to the coefficients of the relevant criterion, we will come back to a percentage of the whole information, the same way we did in the case of AISI 304L.

	Fact3 bis
$F_{p\ av}$	90.87%
$F_{pn\ av}$	-81.82%
R_{av}	90.95%

Tab.4.10. Percentage of the implication of every variable in Fact3

However, the factor 3 bis, like the factor 3 represents 88.84% of the information as it is associated to the third eigenvalue (cf. fig.4.7). Multiplying every value of the factor 3 bis by 0.888, we will get the percentage of the total information included in every variable. (cf. tab.5.4)

	% total info
$F_{p\ av}$	80,73%
$F_{pn\ av}$	-72,69%
R_{av}	80,80%

Tab.4.11. Percentage of the implication of every variable in the total information

Now, the expression of the RC becomes:

$$RC= 0.807*F_{p\ av} -0.727*F_{pn\ av} +0.808*R_{av}$$

Let's check that maximum flank wear can effectively be characterized by this criterion. For that, we will just see the correlation coefficient relating RC to $V_{B\ max}$ corresponding to each preparation:

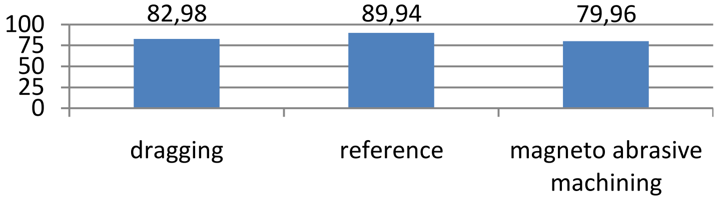


Fig.4.10. Correlation between the RC and $V_{B\ max}$ (for 4140 QT)

As we see in the fig.4.10, correlation between RC and $V_{B\ max}$ is good.

d) From the relevant criterion to wear law

We tried to find the affine relation between RC and $V_{B\ max}$ for each edge preparation. The statistical treatment led to the results below:

Preparation	Mathematical relation between RC and VB max
Dragging	VB max = f(RC) = 0,6610*RC + 0,9972
Reference	VB max = f(RC) = 0,3644*RC - 0,0953
Magneto abrasive machining	VB max = f(RC) = 0,3461*RC - 0,2850

Tab.4.12. Mathematical relation between the RC and $V_{B\ max}$

Basing on the equations found, $V_{B\ max}$ evolution graphs obtained by the RC are compared with graphs obtained by measurements. In this case, values are returned from reduced form to normal form.

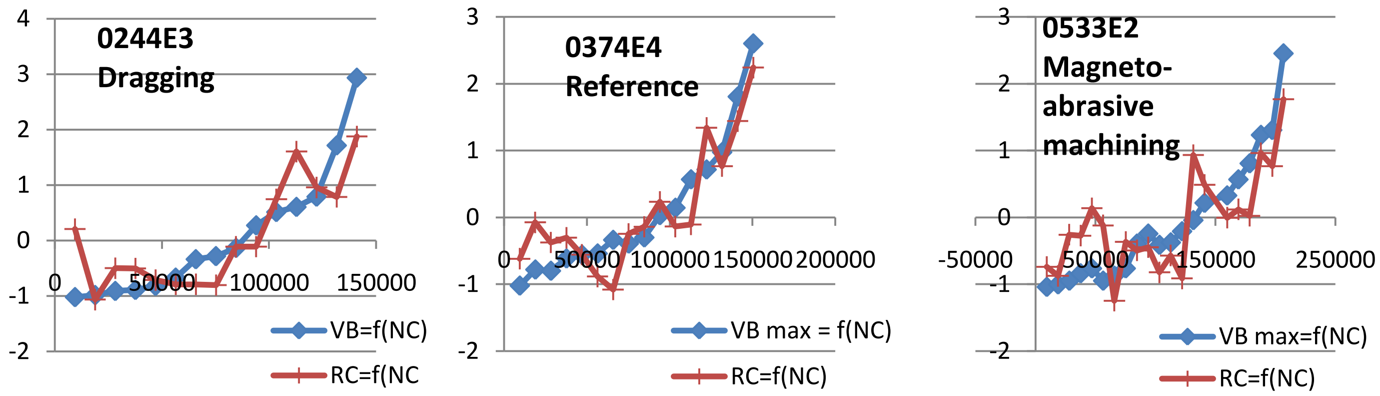


Fig.4.11. $V_{B\ max}$ evolution ($V_{B\ max}$ [µm]=f(NC [rev])) obtained by measurement and by RC

As we see in fig.4.11, in case of the reference, curves collapse with a correlation of 89.94%. However, it is not the case of all preparations. Although the correlation is high, the error percent is still high for edges prepared by magneto-abrasive machining (17.79%) and dragging (33.14%). Here, the best predictable behavior is that of the reference, and the worst is the one obtained by dragging.

4.4. Synthesis

Basing on the PCA, is noticeable that inserts behavior is much different from milling one material to another. In fact, the table below enhances those differences:

	AISI 304L	AISI 4140 QT
Variables best correlating with $V_{B\,max}$	EAA_{max}, $F_{z\,max}$ and R_{max}	$F_{p\,av}$, $F_{np\,av}$ and R_{av}
Average correlation % between RC and $V_{B\,max}$	95.93%	84.29%
Best predictable behavior basing on cutting effort components	Brushing	Reference
Worst predictable behavior basing on cutting effort components	Reference	Magneto-abrasive machining

Tab.4.13. PCA synthesis

We notice that in the case of milling the stainless steel 304L, variables best correlated with maximum flank wear land are calculated in the global cylindrical reference relative to the tool axis, whereas in the case of 4140 QT, variables correlating with $V_{B\,max}$ are calculated in the local reference related to the cutting edge. In fact, in case of the low alloy steel, wear is essentially abrasive, and we expected that friction would have a direct influence on flank wear, manifested in the cutting effort components responsible of the friction between the chip and the insert, in contrary with the case of the stainless steel, where wear is adhesive (proven by the built up edge), and there is chipping. It is then expectable that the local components wouldn't have much influence on flank wear.

However, concerning the average correlation between the relevant criterion and $V_{B\,max}$, 304L measurements have shown the best correlations. Indeed, thanks to the spallings and their quick evolution, the cutting effort and its deviation relatively to the tool axis have increased in a way easily detectable by measurements. On the other side, in the case of the low alloy steel where abrasive wear evolution is too slow, and so the cutting effort components, it was difficult to detect the correlation, especially that the friction has dramatically raised the temperature in the tool-chip surface while wear increases, in a way that we were obliged to stop the trials before the $V_{B\,max}$ limit of 300µm.

Conclusion

In this chapter, a mechanical analysis of the down milling operation was given in order to make a statistical analysis affording us with the relationship between maximum flank wear land and the cutting effort components seen relatively to different references. We realized that flank wear behavior differs from a preparation to another and from a material to another, which makes the choice of a preparation method considering the cutting effort closely tied to its application.

Conclusion & Perspectives

During this project, flank wear resistance of eight inserts is presented. One is the referential insert with no edge preparation; the seven other are prepared respectively by abrasive blasting, honing, abrasive flow machining, dragging, magneto-abrasive machining, brushing, and laser machining. Those SEAN 1203AFTN-M14 inserts are used to down mill the stainless steel AISI 304L and the hardened low alloy steel AISI 4140 QT. Starting with a comprehensive analysis of insert profiles, a best solution for precisely reconstructing the cutting edges was discussed, the process of rough face milling was explained with details including all the working parameters, and then flank wear trials results were revealed in wear charts as well. Then, basing on profile information, the relationship between edge preparation methods and tool life is interpreted. Finally, besides the inserts geometry and wear evolution, we were interested in the correlation between cutting forces and flank wear behavior. So, a mechanical and statistical analysis was led to reveal laws relating cutting force components to the flank wear.

Basing on the results and their analysis, we admit that there is an optimal cutting radius guaranteeing the lowest tool failure. Indeed, that optimization depends on machined material and cutting parameters. So, there is a coupling between the choice of the best preparation method and work materials. However, we have no clear data about the influence of the edge preparation on the insert material structure and state, that we didn’t have the opportunity to quantify it, besides we won’t generalize our analysis in the case of AISI 4140 QT milling, that the results of the experimented edges were not confirmed by a second trial, besides there are some untested preparations due to lack of time. So, on one side, we can confirm our conclusion concerning stainless steel milling, but on the other side, we still have some reserves concerning the quenched and tempered low alloy steel milling.

Moreover, we have noticed that there is a correlation between $V_{B\ max}$ and cutting effort components evolution while milling (with a rake higher than 84% in both cases of milling the stainless steel and the quenched and tempered low alloy steel). However, their behavior is different from a preparation to another and from a work material to another one.

As perspectives, we hope, in the days following this project achievement, wear charts and wear laws of the five remaining preparations unused in AISI 4140 QT milling will be established, and so, we will be able to compare all the results with those of the other laboratories participating in the same Robin Test.

This project opens a discussion concerning the parameters varying with the chosen cutting edge preparation methods: Besides the geometrical impact manifested in r_β variation, are there other parameters affected by these preparations (the cutting edge internal stresses for example)?

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